

Notes on the “pirate” dedisperser: detrending

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1 Time detrending algorithm 1: local polynomial subtraction

Upstream of the tree gridding kernel (*dedispersion.pdf*, “Tree gridding kernel”), each frequency channel is a 1-d intensity time series which must be *detrended*: we subtract a slowly varying baseline (gain drift, sky rotation, imperfectly calibrated bandpass) while leaving short-timescale structure – in particular FRBs – intact.

We have two detrenders for this problem. The one described here is a masked, adaptively centered *moving local polynomial fit*, evaluated by a van Herk block decomposition over a moment monoid. The second, in §2, is a fixed-lag Kalman filter. They are independent – neither section depends on the other – and the intent is to run both and compare them.

1.1 Problem statement

The input is a float array $d[t]$ and a boolean mask $m[t] \in \{0, 1\}$, independently per (beam, frequency) pair. A sample is *valid* if $m[t] = 1$ and junk (RFI-flagged, dropped packet, ...) if $m[t] = 0$. We model the valid samples as

$$d[t] = f[t] + \epsilon[t], \quad (1.1)$$

where $f[t]$ is a smooth trend and $\epsilon[t]$ is white Gaussian noise with zero mean and t -independent variance σ^2 . The detrender computes an estimate $\hat{f}[t]$ and replaces $d[t] \rightarrow d[t] - \hat{f}[t]$ at valid samples.

Outputs. Three per-sample arrays, of which only the first is obvious.

The detrended residual is $r[t] = d[t] - \hat{f}[t]$. Alongside it the detrender emits an *expanded* mask $m_{\text{out}}[t]$: a sample is dropped either because its input sample was masked, or because its window turned out to be too ill-conditioned to fit (§1.2). Where m_{out} is false the residual is meaningless and is set to zero. So the detrender does modify the mask, and a consumer must use m_{out} rather than the mask it supplied.

The third is the conditioning statistic $r_{\min}[t]$ of (1.9), which is what the mask expansion thresholds on.

The leverage, and why it is not one of them. For a *fixed mask* the estimator is linear in the data, so we may write $\hat{f} = Hd$ for a $T \times T$ smoothing matrix H (which depends on the mask but not on d). The *leverage* is its diagonal,

$$\lambda[t] \equiv H_{tt} = \frac{\partial \hat{f}[t]}{\partial d[t]}, \quad (1.2)$$

the sensitivity of the trend estimate at t to the datum at t . Equivalently,

$$\text{Var}(\hat{f}[t]) = \sigma^2 \lambda[t], \quad \text{Var}(r[t]) = \sigma^2 (1 - \lambda[t]), \quad (1.3)$$

exactly, on every surviving sample where the mask is locally constant. So λ is simultaneously the variance of the trend estimate in units of σ^2 , the fraction of the variance the detrender removes, and a measure of how well determined the window is. $\text{Var}(r[t])$ is therefore *not* t -independent even though $\text{Var}(\epsilon[t])$ is: it varies with the local mask, dropping sharply near mask edges and in sparsely populated windows. For a fully valid window $\lambda = 9/8W = 4.4 \times 10^{-3}$, but under adversarial masking λ ranges over several decades and approaches 1, where $\text{Var}(r) \rightarrow 0$ and the residual carries no information at all.

None of that is computed. It would matter if the peak-finder’s normalization to sigmas (*dedispersion.pdf*, “Peak-finding”) divided by $\sigma\sqrt{1 - \lambda[t]}$ rather than by σ , but there is no path by which a per-sample quantity could reach it: the peak-finding weights are built once at startup (*variance_map.pdf*, “Analytic variances from sparse tiles”) from per-zone input variances supplied by the X-engine metadata. Two things make that acceptable rather than merely expedient. Where the mask is locally constant λ is the known constant $9/8W$, i.e. a uniform 0.22% rescaling of σ , degenerate with the overall normalization; and the *variation* about it is confined to a small tail – measured over 2.4×10^5 samples at $W = 512$, 0.07% of surviving samples have $\lambda > 0.1$, which perturbs the variance of a sum over thousands of channels by $O(10^{-4})$. If a correction is ever wanted, the bulk of it is that single analytic constant, not a per-sample array.

A separate question is whether to cut on λ as well as on $r_{\min}$: they answer different questions, and neither subsumes the other. $r_{\min}$ asks whether the *fit* is well posed; λ asks whether the *residual* carries information. Measured over the same samples, every sample with $r_{\min} < \epsilon$ had $\lambda \geq 0.1$, so mask expansion never discards a sample that λ would call healthy – but the converse fails. Those extra samples are well conditioned and numerically accurate, and merely statistically degenerate: $N_v = 3$ with $n = 2$ determines a parabola exactly, so G is perfectly conditioned while $\lambda = 1$ and $\text{Var}(r) = 0$. That decision has been taken and is *not* to cut: high leverage is not a problem for the downstream code, whereas numerical instability is, so the detrender expands the mask on $r_{\min}$ only.

The detrender runs *incrementally*. On the i -th call it is handed a buffer containing chunk i (samples $iT_c \leq t < (i+1)T_c$) together with W samples of pre- and postpadding. Samples $t < iT_c$ have already been detrended and may not be modified, so each output sample is committed once and forever. Parameters:

symbol	meaning	value
n	polynomial degree	2
W	window half-width	256
$2W+1$	window length (odd, symmetric)	513
$B = 2W$	scan block length	512
T_c	time samples per chunk	2048
	buffer length $T_c + 2W$	2560
	scan blocks per chunk, $T_c/B + 1$	5
ϵ	mask-expansion threshold on $r_{\min}$	10^{-3}
	arithmetic	float32

The values are those the kernel is compiled at (`include/pirate/Detrender.hpp`); everything below is written for general (n, W, T_c) and the numpy reference is parameterized by them. An earlier draft of this section used $W = 512$, and the *measured* numbers quoted below were taken there unless stated otherwise; the analytic ones have been recomputed at $W = 256$.

1.2 The estimator

Fix an output time t and let $S_t = \{t - W, \dots, t + W\}$ be its window. Define the valid count and the *mask-weighted centroid*

$$N_v = \sum_{u \in S_t} m[u], \quad c = \frac{1}{N_v} \sum_{u \in S_t} m[u] u, \quad x_u = \frac{u - c}{W}. \quad (1.4)$$

Let $\hat{P}$ be the degree- n polynomial in x minimizing $\sum_{u \in S_t} m[u] (d[u] - P(x_u))^2$, and set

$$\hat{f}[t] = \hat{P}(x_0), \quad x_0 = -c/W, \quad (1.5)$$

i.e. the fit evaluated back at the window *center* $u = t$. Both the choice of origin (the centroid) and the choice of length unit (W) matter, but in opposite ways: the length unit is a diagonal rescaling of the basis and is numerically irrelevant, whereas anchoring the expansion at the centroid rather than at the window center is what keeps the normal equations well conditioned when the mask leaves only a narrow, off-center sliver of valid samples.

Writing $P(x) = \sum_{j=0}^n a_j x^j$, the normal equations are $Ga = U$ with

$$G_{jl} = S_{j+l}, \quad S_r = \sum_{u \in S_t} m[u] x_u^r, \quad U_r = \sum_{u \in S_t} (md)[u] x_u^r, \quad (1.6)$$

so G is the Hankel matrix of the mask moments and needs no assembly beyond indexing. By construction

$S_1 = 0$, hence $G_{01} = 0$. For $n = 2$ this is

$$G = \begin{pmatrix} S_0 & 0 & S_2 \\ 0 & S_2 & S_3 \\ S_2 & S_3 & S_4 \end{pmatrix}, \quad U = (U_0, U_1, U_2)^T. \quad (1.7)$$

Cholesky, and the conditioning statistic. G is singular whenever $N_v \leq n$, and ill-conditioned whenever the valid samples cannot pin down the higher-order shape (two surviving clumps, for instance). Rather than regularize, we detect this and drop the sample. Factor $G = LL^T$ in the usual way; written out for $n = 2$,

$$\begin{aligned} p_0 &= S_0, & L_{00} &= \sqrt{p_0}, & L_{10} &= 0, & L_{20} &= S_2/L_{00}, \\ p_1 &= S_2, & L_{11} &= \sqrt{p_1}, & L_{21} &= S_3/L_{11}, \\ p_2 &= S_4 - L_{20}^2 - L_{21}^2, & L_{22} &= \sqrt{p_2}, \end{aligned} \quad (1.8)$$

where in general $p_i = G_{ii} - \sum_{k < i} L_{ik}^2$. Because $G_{01} = 0$ we get $p_0 = G_{00}$ and $p_1 = G_{11}$ exactly, so the pivots are in natural (level, slope, curvature) order and $p_i/G_{ii} \in [0, 1]$ is the fraction of order- i information surviving after the lower orders are projected out. Define

$$r_{\min}[t] \equiv \min_i \frac{p_i}{G_{ii}}, \quad r_{\min} \equiv 0 \text{ if any } G_{ii} = 0, \quad (1.9)$$

the smallest relative pivot, so that the equilibrated condition number of G is roughly $1/r_{\min}$. The detrender then **masks the output sample when** $r_{\min} < \epsilon$, with $\epsilon = 10^{-3}$.

Three remarks. First, $r_{\min}$ has to look at the *pivots* and not at the diagonal: $N_v = 2$ with the two samples at $\pm x$ gives $G_{ii} > 0$ for every i and yet $p_2 = 0$ exactly, since G has rank 2. Second, since a rank- k G has $p_i = 0$ for $i \geq k$, this masks exactly the windows that cannot determine a degree- n fit – $N_v \leq n$ always, and more besides. It is a conditioning test and not a sample count: $N_v = 3$ spread across the window survives, while $N_v = 300$ in two tight clumps does not. Third, and this is the reason for choosing $r_{\min}$ as the threshold quantity,

$$r_{\min} \geq \epsilon \implies \text{cond}(\text{equilibrated } G) \leq O(1/\epsilon) = O(10^3), \quad (1.10)$$

so on every sample we keep, the relative error of the solve is bounded by $\epsilon_{\text{mach}}/\epsilon \approx 1.2 \times 10^{-4}$ *independently of the mask*. That bound is what makes the float32 implementation safe (§1.5).

Masking rather than regularizing costs nothing on the samples we keep, and buys exactness. Any multiplicative pivot floor $p_i \rightarrow \max(p_i, \epsilon G_{ii})$ would be *inert* wherever $r_{\min} \geq \epsilon$, so the two agree on every surviving sample and differ only in the disposition of the rest. Dropping them instead of shrinking their fits means no surviving sample carries any shrinkage bias, so polynomial reproduction and the identities (1.3) hold exactly rather than approximately. The price is a mask-expansion rate measured at 5×10^{-4} over the randomized-mask test suite.

One implementation detail: p_i can be zero, or slightly negative through cancellation, on a sample we are about to mask, and every sample is evaluated branch-free. Replace p_i by $\max(p_i, \mu)$ with $\mu = 10^{-30}$ before the square root, purely to keep NaNs out of the batch. μ is a guard, not a tuning parameter: it is inert wherever $r_{\min} \geq \epsilon$.

Solve and evaluate. With $w = (1, x_0, \dots, x_0^n)^T$,

$$\hat{f}[t] = w^T a, \quad LL^T a = U, \quad (1.11)$$

by forward then back substitution.

Leverage. The leverage (1.2) of this estimator is

$$\lambda[t] = \|z\|^2 = w^T G^{-1} w, \quad Lz = w, \quad (1.12)$$

so one extra $O(n^2)$ forward substitution would give it, exactly on every surviving sample since there is no regularizer. It is not computed (§1.1); the expression is recorded because the response statements below are statements about it. For a fully valid window $\lambda = 9/8W$, so the detrender removes 0.44% of the variance and $\text{sd}(\hat{f}) = 1.06 \sigma / \sqrt{W} = 0.066 \sigma$ at $W = 256$.

Response. For a fully valid window the estimator is a fixed FIR smoother with $1 - H(\omega) = O(\omega^4)$ at low frequency and a smoothing timescale $\sim W$. Two consequences worth keeping in view. A pulse of width $w \ll W$ loses a fraction $\approx 1.125 w/W$ of its flux into the trend (28% at $w = 64$, $W = 256$), so W must stay well above the widest searched boxcar $W_{\max}$. And H is a truncated polynomial kernel, so $1 - H$ has stopband ripple: at $n = 2$ the first and largest lobe is $H = -0.242$ at $\omega \approx 2.2\pi/W$ (exactly, $W\omega = 6.988$ in the large- W limit), where the detrender *amplifies* by 24% rather than attenuating. This is a property of the estimator, not of the implementation. Both numbers come from the closed form: the equivalent kernel is $h[s] = a + bs^2$ with $a = 3(3P - 1)/(N(4P - 3))$ and $b = -15/(N(4P - 3))$, where $N = 2W + 1$ and $P = W(W + 1)$, so $H = aD_W + bS_W$ with D_W the Dirichlet kernel $\sin(N\omega/2)/\sin(\omega/2)$ and $S_W = -D_W''$; as $W \rightarrow \infty$ this tends to $H(x) = \frac{15}{2}(x^{-3} \sin x - x^{-2} \cos x) - \frac{3}{2}x^{-1} \sin x$ with $x = W\omega$, which is $1 - x^4/280 + O(x^6)$ at low frequency. ($n = 1$ is the boxcar, whose ripple is 0.217 at $\omega \approx 1.4\pi/W$.)

1.3 The moment monoid

Everything in (1.6) is computed by combining partial sums, so define for any sample set S

$$\mathcal{M}(S) = (N_v, c, S_2, \dots, S_{2n}, U_0, \dots, U_n), \quad (1.13)$$

each moment taken about that set's own centroid c , with $S_1 \equiv 0$ omitted. This is $3n + 2$ numbers (8 at $n = 2$). Change of origin acts by the Pascal matrix

$$(T_\delta v)_r = \sum_{j \leq r} \binom{r}{j} \delta^{r-j} v_j, \quad T_\delta T_{\delta'} = T_{\delta + \delta'}, \quad (1.14)$$

and for disjoint S, S' the merge $\mathcal{M}(S) \oplus \mathcal{M}(S') = \mathcal{M}(S \cup S')$ is

$$N = N_v + N'_v, \quad f = N'_v/N, \quad \Delta = (c' - c)/W, \quad c'' = c + f\Delta W, \quad \delta = -f\Delta, \quad \delta' = (1 - f)\Delta, \quad (1.15)$$

followed by applying T_δ and $T_{\delta'}$ to the two moment vectors and adding componentwise. The centroid update is written in this form (rather than $(N_v c + N'_v c')/N$) to avoid forming the large product. Note $|\delta| + |\delta'| = |\Delta|$ exactly, so the total shift is the centroid separation and no scheme can do better. The $r = 1$ rows need not be computed: they cancel to zero by the definition of c'' , which is both a flop saving and a good debug assertion.

The operation is associative and commutative (the weighted mean is associative; moments about a common origin are set-additive), with identity $N_v = 0$, all moments zero, and c arbitrary. So $(\cdot, \oplus)$ is a commutative monoid and supports a parallel scan.

Empty sets. c is undefined when $N_v = 0$, and a tree scan performs many merges involving empty aggregates (any fully masked sub-block). Two rules make this safe and branch-free: initialize c for an empty set to a *finite* nominal value (its sub-block center), and guard the divide, $f := 0$ when $N = 0$. Then merging empty with non-empty gives $f = 1$, $\delta' = 0$, $c'' = c'$, which is correct, and merging two empties propagates zeros. Getting this wrong injects NaNs into every window touching a masked block.

1.4 Block decomposition and scans

Partition the buffer into blocks of length $B = 2W = 512$, anchored at (chunk start $-W$); since T_c is required to be a multiple of B (here $T_c = 4B$) this lattice is the same for every chunk, each chunk needs exactly $T_c/B + 1 = 5$ blocks, and the last block of chunk i is the first block of chunk $i + 1$.

A window has length $B + 1 = 513$, so it spans *exactly* two adjacent blocks for every alignment: an interval $[q, q + B]$ meets blocks $\lfloor q/B \rfloor$ and $\lfloor q/B \rfloor + 1$ and no others. Writing $p = q \bmod B$,

$$\mathcal{M}(S_t) = \text{Suff}_b[p] \oplus \text{Pref}_{b+1}[p], \quad \underbrace{(B - p)}_{\text{offsets } p \dots B-1} + \underbrace{(p + 1)}_{\text{offsets } 0 \dots p} = B + 1, \quad (1.16)$$

where Pref and Suff are inclusive prefix and suffix scans of the monoid within a block. There is no special case at either end: $p = 0$ takes all of block b plus one sample of block $b + 1$, and $p = B - 1$ the reverse. An odd window over even blocks is what makes this work, and it also keeps the fit exactly symmetric (so $S_{\text{odd}} = 0$ exactly on a full window) and puts the evaluation point on a real sample rather than half way between two.

The scans must be trees, and must carry the centroid. Two requirements, both load-bearing:

- Build Pref and Suff with a tree scan, not a sequential one, so that each output depends on its leaves through $O(\log B)$ merges rather than $O(B)$. Which tree does not matter – the load-bearing property is the depth, and both implementations use Hillis-Steele (the numpy reference because work-efficiency is irrelevant there, the kernel because it is what a warp shuffle scan does) where a work-efficient Brelloch scan would do equally well. Measured worst-case moment error over a full block in float32, against a float64 reference: 3.8×10^{-5} for the tree versus 4.0×10^{-4} sequential, a factor of 10. (An earlier draft quoted 170; that came from a pure *reduction*, which returns only the root value, whereas a prefix scan must be judged by the worst of all B outputs.) It also matches the structure of the GPU implementation (warp-level shuffle scan, then a block-level tree), which is what makes the CPU reference usable for validating it.
- Every partial aggregate must carry its own centroid, per (1.15). Accumulating the scans about a fixed block center and shifting to the centroid at the end is much cheaper and completely wrong: a prefix holding a cluster of width w at distance D from the block center has moments $\sim N_v(D/W)^r$, and the final shift must cancel these down to $\sim N_v(w/W)^r$. In the worst measured case this destroys every digit (100% relative error in $\hat{f}$) where centroid-carrying scans give 5×10^{-7} .

Origin travel. Because the scans restart at every block, the accumulated origin travel along any path is bounded by the block width, $|\Delta| \leq 2$ in units of W , and the associated error amplification by $(1 + |\Delta|)^{2n}$. This bound is structural – it does not depend on tuning a re-originating interval – and it is the reason a sliding accumulator carried across a whole chunk is not an option (there $|\Delta|$ reaches $T_c/W = 4$, and the amplification 1.6×10^4).

1.5 Numerical requirements

Everything runs in float32. The budget follows from (1.10): on a surviving sample the equilibrated condition number is at most $O(1/\epsilon)$, so the relative error of the solve is at most $\epsilon_{\text{mach}}/\epsilon \approx 1.2 \times 10^{-4}$. Measured worst case over the randomized-mask test suite is $8.7 \times 10^{-5}\sigma$, consistent with that bound, and far below $\hat{f}$'s own statistical error of 0.047σ . The bound holds *independently of the mask*, which is the property that matters: it is what stops the accuracy from degrading as the mask distribution is made more adversarial. Four requirements keep us there.

- **Subtract a constant offset first.** Intensity data is total power, so $d[t]$ carries a large positive mean (system temperature) while everything of interest – the trend and the pulse alike – lives at the σ level on top of it. Every accumulator in (1.6) then sums terms of magnitude $|d|$, and floating-point rounding is proportional to the accumulator's magnitude rather than to σ , so the effective precision of U_r is degraded by the factor $|d|/\sigma$. Removing the offset costs nothing: the polynomial basis contains the constant function, so the fit is exactly translation-equivariant, and replacing d by $d - \kappa$ shifts $\hat{f}$ by exactly κ and leaves the residual $d - \hat{f}$ unchanged. Nothing needs adding back, and κ needs no

accuracy at all: any value is exact, and it affects only rounding. What it must not be is *stale*. Take κ to be the masked mean of the buffer in hand, not of the previous chunk: an inherited value collapses to zero whenever the previous chunk happened to contain no valid samples, which silently disables the subtraction for the current one. Measured, that costs $5 \times 10^{-3}\sigma$ in float32 at an offset of $10^4\sigma$, against $1 \times 10^{-7}\sigma$ when κ comes from the current buffer. Using the current buffer is also cheaper (there is no state to carry) and makes the per-chunk computation a pure function of its arguments. The residual error scales as $3.4\epsilon_{\text{mach}}|d - \kappa|$, so the requirement is $|d - \kappa| \leq 10^3\sigma$. Note this bounds the *spread* of d within one buffer rather than its absolute level: no single additive constant helps when a large step falls inside a window, where a step of $10^5\sigma$ still costs $2 \times 10^{-2}\sigma$. Seen less numerically, the offset *is* the zeroth-order piece of the trend: this just splits the trend into a crude constant handled once per chunk plus the local polynomial handled per sample.

- **Rebuild per chunk.** Moment state is never carried across calls; each chunk’s scans are built from scratch. (The shared block between consecutive chunks may be cached, but as a recomputation-avoidance optimization only.)
- **Monomials in x , no orthogonalization.** $\text{cond}(G)$ looks alarming ($\sim 10^{15}$ in raw sample units) but is a pure diagonal-scaling artifact; Cholesky is equivariant under diagonal scaling and the relevant quantity, the condition number of the equilibrated matrix $G_{jl}/\sqrt{G_{jj}G_{ll}}$, is 23. This is the standard equilibration result (van der Sluis; see Higham, *Accuracy and Stability of Numerical Algorithms*): the error bound for Cholesky depends on the equilibrated condition number, not on $\text{cond}(G)$. Legendre or Gram polynomials would reduce that to $O(1)$, which buys nothing, and would cost the Hankel structure of (1.6).
- **ϵ must clear the rounding noise in $r_{\min}$.** Mask expansion is a threshold decision, so it is only well defined if ϵ sits well above the precision to which $r_{\min}$ itself is computed. Measured worst-case disagreement between a float32 and a float64 run is 1.6×10^{-5} , so $\epsilon = 10^{-3}$ has a margin of $60\times$. Note this is the *only* place float32 precision feeds back into a discrete decision; everywhere else it perturbs a continuous quantity.

Two things follow that are worth stating positively. $r_{\min}$ is a ratio of a pivot to a diagonal entry, hence covariant under diagonal rescaling of the basis, so the threshold ϵ needs no calibration against σ , against the trend amplitude, or against the effective window width – unlike an absolute ridge, which has to be recalibrated as the mask narrows. And the masking decision depends only on G , hence only on the mask, so $\hat{f}$ remains linear in d for a fixed mask; the transfer function, the identities (1.3), and Monte-Carlo noise propagation all remain valid.

1.6 Cost

Per output sample, at $n = 2$, counting FMAs. A merge (1.15) is ≈ 60 flops: two Pascal shifts (15 for the mask part, 6 for the data part, each), 7 adds to combine, and 11 for the centroid update. A work-efficient tree scan costs ≈ 2 merges per element and we need both a prefix and a suffix scan, over 2560 input samples per 2048 outputs:

	flops/output
prefix + suffix tree scans (4 merges/input $\times 1.25$)	300
window merge (1.16), one per output	60
Cholesky, two solves, evaluation	35
total	≈ 400

The tree scan roughly doubles the scan term relative to a sequential scan; we pay it for the factor of 10 in accuracy and for GPU parity. The cost is *independent of the mask*, which is the main reason for choosing

this decomposition: in a real-time system the latency budget must be met in the worst case, and RFI is bursty, so a scheme whose cost degrades under heavy masking degrades for many channels at once.

At $n = 3$ the merge grows to ≈ 115 flops and the total to ≈ 715 , for an answer that is *identical* wherever the window is fully valid (with $S_{\text{odd}} = 0$ the Gram matrix is checkerboard and a_0 depends only on the even block). Degree 3 differs from degree 2 only where the mask is asymmetric within the window, so we start at $n = 2$ and measure whether it matters near mask edges.

One consequence of $n = 3$ is worth flagging in advance, because it is structural rather than a matter of flops: **the kernel’s division-free mask test does not survive it**. At $n = 2$, $G_{01} = 0$ gives $p_0 = G_{00}$ and $p_1 = G_{11}$ exactly, so $r_0 = r_1 = 1$ and $r_{\min}$ collapses to the single ratio $p_2/G_{22} = \det G / (G_{00}G_{11}G_{22})$. That is what lets `DetrenderLps1d.cu` evaluate the fit through the adjugate and test `det >= eps*N*A*C` with no square roots. At $n = 3$ we still have $r_0 = r_1 = 1$, but r_2 and r_3 both bind, and $\min(p_2/G_{22}, p_3/G_{33})$ cannot be recovered from $\det G$ alone – it needs the leading principal minors, i.e. the factorization. So $n = 3$ means an unrolled Cholesky in the kernel rather than an adjugate. That costs nothing in accuracy: measured over the randomized-mask suite at $n = 2$, $W = 256$, restricted to the windows the detrender actually keeps ($m[t] = 1$ and $r_{\min} \geq \epsilon$), the two solves agree to within a factor of two in the worst case ($2.8 \times 10^{-7}\sigma$ for Cholesky against $5.1 \times 10^{-7}\sigma$ for the adjugate, with equal medians at $5 \times 10^{-9}\sigma$), and both sit two orders of magnitude below the $8.7 \times 10^{-5}\sigma$ the full pipeline achieves. The float32 budget is spent in the moment scans, not in the solve.

Storage for the flat scheme is $3n+2$ floats $\times B \times$ two scans, i.e. 32 KB per block at $n = 2$. A two-level variant avoids this, and is what the kernel does: within sub-blocks of $B/32 = 16$ samples the span is only $16/W = 0.06$ in normalized units, so no adaptive centering is needed there and the sub-block partials are pure accumulation maintainable in registers; only the 32 sub-block aggregates per block carry centroids, which is 1 KB. Since the mask is per (beam, frequency) and there are thousands of such pairs, the natural mapping is one thread per pair walking time sequentially, with the time axis fastest in memory.

1.7 Validation

Two things make this unusually testable. A float64, direct-summation implementation of §1.2 – no blocks, no scans, one $O(W)$ sum per output – is a few dozen lines and serves as a reference for the block/scan machinery. And the estimator has an exact fixed point: fed a degree- $\leq n$ polynomial with no noise, it must return *identically zero* at every surviving sample. The second is the sharper test, because the expected answer is known analytically, with no reference implementation and no statistical tolerance – whatever residual appears is pure accumulated roundoff. It is also what mask expansion made unconditional: with a regularizer there would be a shrunk-fit carve-out to exclude.

Masks are randomized rather than enumerated. Each row of a test mask independently draws a type (all-valid with probability 1/2, then Bernoulli at a random rate, long gaps, one-sided, narrow off-center clusters, two clumps, periodic dropouts, fully masked scan blocks, sparse lattices, all-masked), randomizes that type’s parameters, and is then perturbed by a random set of fully-masked and fully-valid subintervals whose lengths follow a power law. Rows are independent, which is what exercises the spectator axis.

The checks worth keeping as regressions:

- **Monoid algebra:** (1.15) against direct moments, associativity, commutativity, identity, $S_1 \equiv 0$, and the empty-set rule of §1.3 (the NaN trap).
- **Block decomposition:** (1.16) against direct moments for *every* offset p , which is what pins the odd-window-over-even-block arithmetic at both endpoints.
- **Polynomial exactness**, as above, at both precisions and over several chunks. Running it through the chunked path is also what covers chunk stitching: a wrong block anchor or an off-by-one in the buffer slicing shows up as a nonzero residual, and a seam shows up as structure at multiples of T_c . This subsumes a separate seam test.

- **Degenerate windows:** $N_v = 0, 1, 2$ must all be mask-expanded away, with a positive control that $N_v = n + 1$ spread across the window survives.
- **Projection identity:** build the equivalent kernel κ by feeding unit impulses through the full path and check $\kappa[0] = \sum_s \kappa[s]^2$. Both sides equal λ by (1.12), but the identity is checked from the kernel alone, so it needs nothing the solve does not already produce. Doing this by impulse response rather than by pushing noise through and measuring $\text{Var}(r)$ is exact, needs no statistical tolerance, and subsumes a Monte-Carlo variance test.
- **Kernel at zero lag:** on a full window, $\kappa[0] = (G^{-1})_{00}$ for the exact discrete Gram, which approaches $9/8W$ as W grows. This is the sharpest check here, since the expected answer is analytic – no reference implementation and no statistical tolerance – and one impulse suffices.
- **Against the float64 reference,** over the randomized masks, comparing r , m_{out} and r_{min} .
- **Float32 versus float64,** the whole pipeline in both. Run the two at *different* ϵ (10^{-3} against 10^{-6}) and compare on the intersection of their masks: with matched ϵ the two runs make identical masking decisions, so the test never exercises a sample whose r_{min} roundoff was large enough to move it across the threshold. Also track the cumulative mask-expansion rate, which is the diagnostic for a systematic bias in float32's r_{min} – that would show up as an elevated expansion rate rather than as a residual discrepancy.

Two properties are checked analytically rather than in code, because they are properties of the estimator and cannot regress when the implementation changes: the $O(\omega^4)$ rolloff of $1 - H$ (which is the same statement as polynomial exactness, in the other domain) and the $1.125 w/W$ flux loss.

2 Time detrending algorithm 2: Kalman filter

This is the second detrender. It solves the problem of §1.1 – same inputs $d[t]$ and $m[t]$, same requirement to run incrementally, same four output arrays – and differs from §1 in what it assumes about the trend. There, the trend is whatever a degree- n polynomial can follow across a window of $2W+1$ samples, and the window is what makes the estimate local. Here the trend is a random process with a characteristic timescale, the estimate is a posterior mean, and there is no window at all.

The structural difference in the contract. The estimate committed at time t is

$$\hat{f}_L[t] = E[f[t] \mid d[0..t+L]], \quad (2.1)$$

a *fixed-lag* estimator: it depends on t and the lag L only, and not at all on where the chunk boundaries fall. It is therefore **seam-free by construction** – not to a small tolerance, but exactly. (A seam is a step discontinuity in the committed $\hat{f}$, which is a broadband transient, i.e. precisely what the dedispersion search is built to find. §1 avoids seams by making each output depend only on a window centered on it; this section avoids them by letting the endpoint move with t .) The price is memory: the buffer handed to the i -th call is chunk i plus L samples of postpadding and *no prepadding*, with $O(k^2)$ of filter state carried across the call boundary, so chunks must be processed in order.

2.1 The model

Fix an integer order $k \geq 1$. The trend is a k -fold integrated random walk observed in noise:

$$\begin{aligned} \text{prior: } (\Delta^k f)[t] &= w[t], & w[t] &\sim \mathcal{N}(0, q) \text{ i.i.d.}, \\ \text{likelihood: } d[t] &\sim \mathcal{N}(f[t], \sigma^2) & \text{at valid } t \text{ only,} \end{aligned} \quad (2.2)$$

where $(\Delta f)[t] = f[t+1] - f[t]$. For $k = 1$ the trend is a random walk; for $k = 2$ it has a level and a slope, and the slope wanders.

The point of (2.2) is that it is Markov in the k -dimensional state

$$x[t] = \left(f[t], (\Delta f)[t], \dots, (\Delta^{k-1} f)[t] \right)^T, \quad (2.3)$$

since $x_j[t+1] = x_j[t] + x_{j+1}[t]$ for $j < k-1$ and $x_{k-1}[t+1] = x_{k-1}[t] + w[t]$. Writing N for the superdiagonal shift,

$$x[t+1] = A x[t] + g w[t], \quad A = I + N, \quad g = e_{k-1}, \quad d[t] = e_0^T x[t] + \epsilon[t]. \quad (2.4)$$

A is the Pascal matrix, $(A^s)_{jl} = \binom{s}{l-j}$, and $A^{-1} = I - N + N^2 - \dots$ terminates because N is nilpotent. Both are exact integer matrices in any floating-point type.

Equivalent penalized least squares. With a *diffuse* prior on $x[0]$ – no prior at all – the posterior mode of (2.2) is the minimizer of

$$\chi^2(f) = \sum_t m[t] (d[t] - f[t])^2 + \rho \sum_t (\Delta^k f)[t]^2, \quad \rho = \sigma^2/q, \quad (2.5)$$

whose normal equations are

$$\mathcal{N} \hat{f} = M d, \quad \mathcal{N} \equiv M + \rho D_k^T D_k, \quad M = \text{diag}(m), \quad (2.6)$$

with D_k the k -th difference matrix. Two consequences are used throughout.

$\mathcal{N}$ is positive definite as soon as the array holds k valid samples anywhere: the null space of D_k is the polynomials of degree $< k$, and a nonzero one has at most $k-1$ roots. So there is no per-window rank deficiency to detect and no analogue of the ill-conditioned windows of §1.2; the prior regularizes everything,

and mask handling reduces to “skip the measurement update”, which is exactly correct because a masked sample carries no information.

And the estimator depends on (k, ρ) only, *not* on σ : scaling $d \rightarrow cd$ scales $\hat{f} \rightarrow c\hat{f}$ with ρ fixed. So σ never has to be known or estimated, and every formula below may be written with $\sigma^2 = 1$ and $q = 1/\rho$. What this does assume is that σ is independent of t within a row.

Timescales. Three derived quantities recur:

$$\tau = \rho^{1/2k}, \quad \ell = \frac{\tau}{\sin(\pi/2k)}, \quad c_k = \frac{1}{2k \sin(\pi/2k)}, \quad \ell = 2k c_k \tau. \quad (2.7)$$

τ is the smoothing timescale in samples and is the parameter to think in. ℓ is the correlation length of $\mathcal{N}^{-1}$, whose entries decay as $\exp(-|t - t'|/\ell)$; it is what governs how much lookahead is needed. c_k is the flux-loss constant of (2.16). At $k = 2$, $c_2 = 0.354$ and $\ell = \sqrt{2} \tau$.

2.2 The estimator

Because x is Markov and the data split at t , the posterior in (2.1) factorizes:

$$p(x[t] \mid d[0..t+L]) \propto \underbrace{p(x[t] \mid d[0..t])}_{\text{forward filter}} \cdot \underbrace{p(d[t+1..t+L] \mid x[t])}_{\text{backward likelihood}}. \quad (2.8)$$

Both factors are Gaussian in $x[t]$. Carry each in *information* form – the pair (J, η) standing for $\exp(-\frac{1}{2}x^T J x + \eta^T x)$, so J is a precision and $\eta = J\mu$ – and the combination is pure addition:

$$\boxed{J[t] = J_f[t] + J_b[t], \quad \eta[t] = \eta_f[t] + \eta_b[t], \quad \hat{f}_L[t] = (J[t]^{-1} \eta[t])_0.} \quad (2.9)$$

That is the whole algorithm. Information form rather than the more familiar covariance form for three reasons, each of which is load-bearing: the diffuse initialization is *exactly* $J_f = 0$, with no large-variance fudge, which is what makes the polynomial reproduction of §2.4 exact rather than approximate; the measurement update becomes a pure addition; and the backward factor is a *likelihood* rather than a distribution, so it is allowed to be rank-deficient – which it is whenever $[t+1, t+L]$ holds fewer than k valid samples, a state the covariance form cannot represent at all.

The two recursions. They are mirror images: measure, rank-one downdate, then a congruence by A^{-T} going forwards or A^T going backwards. Forwards, initialize $(J_f, \eta_f) = (0, 0)$ before $t = 0$ and at each t

$$\begin{aligned} \text{measure: } J_f &+= m[t] e_0 e_0^T, & \eta_f &+= m[t] d[t] e_0, \\ \text{predict: } M &= A^{-T} J_f A^{-1}, & \zeta &= A^{-T} \eta_f, & \beta &= g^T M g + 1/q, \\ J_f &\leftarrow M - \frac{M g g^T M}{\beta}, & \eta_f &\leftarrow \zeta - \frac{M g (g^T \zeta)}{\beta}, \end{aligned} \quad (2.10)$$

where the post-measurement, pre-predict value is the $J_f[t]$ of (2.9). Backwards, to summarize $d[t+1..t+L]$ as a likelihood in $x[t]$, initialize $(J_b, \eta_b) = (0, 0)$ at $u = t+L$ and step u down to t , absorbing the sample at $u+1$:

$$\begin{aligned} \tilde{J} &= J_b + m[u+1] e_0 e_0^T, & \tilde{\eta} &= \eta_b + m[u+1] d[u+1] e_0, & \beta &= g^T \tilde{J} g + 1/q, \\ J_b &\leftarrow A^T \left(\tilde{J} - \frac{\tilde{J} g g^T \tilde{J}}{\beta} \right) A, & \eta_b &\leftarrow A^T \left(\tilde{\eta} - \frac{\tilde{J} g (g^T \tilde{\eta})}{\beta} \right). \end{aligned} \quad (2.11)$$

Skipping the update at $m = 0$ is the entire handling of the mask. Note the only division in either recursion is by β , and $g^T(\cdot)g \geq 0$ by positive semidefiniteness, so $\beta \geq 1/q > 0$ always: unlike the moment monoid of §1.3 there is no guarded divide and no empty-set rule anywhere. The downdate is a Schur complement of a

PSD matrix, so it preserves both positive semidefiniteness and rank exactly, and $\text{rank } J[t] = \min(j, k)$ where j is the number of valid samples in $[0, t + L]$.

The backward window slides, and cannot be slid. The forward pass is a running prefix and is computed once. The backward factor is over $[t+1, t+L]$, which moves with t , and the tempting trick – walk it by adding the entering sample and subtracting the departing one, as §1 walks its moment window – does not work. The new sample enters at the *far* end of the chain, which is the bottom of the recursion; and although the single-sample step is invertible, being $\tilde{J} \mapsto (\tilde{J}^{-1} + q gg^T)^{-1}$, the recursion forgets exponentially on scale ℓ , so its inverse amplifies exponentially. The obstruction is exactly the rank-one downgrade, i.e. exactly the process noise: at $q = 0$ it vanishes, the accumulation becomes additive, and J_b degenerates to the masked Gram matrix $\sum_u m[u](A^{u-t})^T e_0 e_0^T A^{u-t}$ of a global polynomial fit – which is (1.6) in the binomial rather than the monomial basis, A^s being the Pascal shift. So the moment monoid is the $q \rightarrow 0$ degeneration of this construction, and the process noise plays the role that the finite window plays there: both exist to localize a fit that would otherwise be global.

The practical consequences are in §2.7.

2.3 Outputs

The same three per-sample arrays as §1.1, so the two detrenders are drop-in comparable.

Residual and expanded mask. Factor $J[t] = LL^T$ and, exactly as in (1.9), take the smallest relative pivot

$$r_{\min}[t] = \min_i \frac{p_i}{J_{ii}} \quad (0 \text{ if any } J_{ii} = 0), \quad m_{\text{out}}[t] = m[t] \wedge (r_{\min}[t] \geq \epsilon), \quad (2.12)$$

with $\epsilon = 10^{-3}$. This inherits the guarantee of §1.2: on a surviving sample the equilibrated condition number is bounded, hence the relative error of the solve is $O(\epsilon_{\text{mach}}/\epsilon)$ *independently of the mask*. Where m_{out} is false, both the residual and $r_{\min}$ are set to zero, so a consumer that forgets to check the mask cannot pick up a meaningless $r_{\min}$. The expansion is a single pass: it is never fed back into J_f , J_b or the carried state, which is what keeps $\hat{f}$ linear in d for a fixed input mask, and hence keeps m_{out} and $r_{\min}$ functions of that mask alone.

Because a rank-deficient J has $p_i = 0$ for $i \geq \text{rank}$, (2.12) subsumes the rank test: “fewer than k valid samples in $[0, t + L]$ ” and every near-degenerate variant of it fall out of the same comparison. Since the fixed-lag prefix reaches back to the start of the stream, this can only fire at the start of a stream (or after a state reset), so unlike §1.2 the expansion rate is not a per-sample property at all: it is a fixed handful of samples per stream, diluted by however long the stream runs.

Leverage. With $\sigma^2 = 1$,

$$\lambda[t] = (J[t]^{-1})_{00} = H_{tt} = \frac{\partial \hat{f}_L[t]}{\partial d[t]}, \quad (2.13)$$

which the same factorization would give for free. Like the local fit’s (1.12) it is not computed (§1.1); the expression is recorded because (2.16) below is a statement about it. That this is the leverage and not merely the posterior variance is worth a line: $\hat{f}_L[t]$ is row t of the *full* smoother of the sub-problem on $[0, t + L]$, and for that sub-problem the posterior precision of f is $\mathcal{N}/\sigma^2$, so $\Sigma = \sigma^2 \mathcal{N}^{-1}$ and comparing with $\hat{f} = \mathcal{N}^{-1} M d$ gives $H = \Sigma M / \sigma^2$, whence $H_{tt} = \Sigma_{tt} / \sigma^2$. The fixed-lag H_L is itself *not* symmetric – each row comes from a different sub-problem – but the diagonal identity survives because each row is a row of a symmetric matrix. Note $0 \leq \lambda \leq 1$ at every valid sample, since $e_t^T \mathcal{N}^{-1} e_t = \max_v v_t^2 / (v^T \mathcal{N} v) \leq \max_v v_t^2 / (v^T M v) = 1$.

Residual variance needs a second number. In §1 the restricted H is a projection, so $\text{Var}(r) = \sigma^2(1 - \lambda)$ exactly and one number does both jobs. A penalized estimator *shrinks* rather than projects, and instead

$$\frac{\text{Var}(r[t])}{\sigma^2} = 1 - 2\lambda[t] + \nu[t], \quad \nu[t] \equiv \frac{\text{Var}(\hat{f}_L[t])}{\sigma^2} = \sum_u m[u] H_{tu}^2 \leq \lambda[t]. \quad (2.14)$$

Neither λ nor ν is computed, so this matters only if the residual variance is ever wanted. It is recorded

because it is the one place where the two detrenders genuinely differ in what a correction would cost: λ would be free from the factorization, whereas ν needs a Lyapunov recursion for $\text{Cov}(\eta_f)$ and $\text{Cov}(\eta_b)$ alongside each information recursion, roughly doubling the cost. Using $1 - \lambda$ in its place overestimates the residual variance, since $\nu \leq \lambda$, and therefore understates significance – wrong, but in the conservative direction, and both vanish together as $\lambda \rightarrow 1$. How much that shortcut costs has been measured: over the randomized-mask suite $\max(\lambda - \nu)$ ran 0.12 to 0.19, so on adversarial masks the discrepancy is real, while in well-sampled stretches $\lambda \approx c_k/\tau$ and $\nu \leq \lambda$ bound it below a percent. The number is quoted here because reproducing it means reinstating both quantities in the dense oracle of §2.8, neither of which that oracle now computes.

2.4 Exact properties

Three, and they are what the implementation is tested against.

Chunk invariance, exactly. The forward recursion is a pure left-to-right sequence and the backward window lies entirely inside the buffer, so the arithmetic performed for output t does not depend on the chunk decomposition at all. Running the stream at any chunk size must agree with a single whole-stream call *bit for bit*. Moreover J_f and J_b are driven by the mask alone – a measurement adds $e_0 e_0^T$ regardless of d – so m_{out} and r_{min} are bit-identical unconditionally, while the residual is bit-identical once the constant offset of §2.7 is held fixed. This is the operational meaning of “seam-free”.

Polynomial reproduction, at two different strengths. Setting $f = P$ in (2.6) leaves the residual $\rho D_k^T D_k P$, and $(D_k^T D_k P)[j] \propto (\Delta^{2k} P)[j - k]$ for $k \leq j \leq T' - k - 1$. Hence:

- **Degree $\leq k - 1$: exact, unconditionally.** Here $f = P$ zeroes *both* terms of (2.5) at once, so it is the unique minimizer for any mask, any array length and any position, and the residual is identically zero. Analytic, with no reference implementation and no statistical tolerance.
- **Degree k to $2k - 1$: exact on the infinite array.** $\Delta^{2k} P = 0$ kills the interior rows but not the k truncated rows at each end, so reproduction fails only near an *array endpoint* – and notably *not* at a mask edge: inside a gap the equation reduces to $D_k^T D_k f = 0$, which P satisfies, so a degree- $(2k-1)$ polynomial continues straight through a gap of any length. The fixed-lag estimator always has an endpoint L away, so in practice the residual is $O(e^{-L/\ell})$ rather than zero. Measured at $k = 2$, $\tau = 32$ on a full mask, the degree-3 residual falls $2.5 \rightarrow 1.7 \times 10^{-1} \rightarrow 2.5 \times 10^{-2} \rightarrow 9.1 \times 10^{-6} \rightarrow 8.5 \times 10^{-13}$ as L/ℓ runs 2, 4, 8, 16, 32, i.e. as $e^{-L/\ell}$ with a slowly growing prefactor.
- **Degree $\geq 2k$: never**, by $O(1)$.

The degenerate window is well defined. When $[0, t + L]$ holds fewer than k valid samples, J is singular; the system stays *consistent* nonetheless, since $\eta \in \text{range}(J)$ by induction from $(0, 0)$, and if $m[t] = 1$ then $e_0 \in \text{range}(J)$ because the measurement at t contributes $e_0 e_0^T$. So $e_0 \perp \ker J$ and $\hat{f}_L[t]$ is uniquely determined even though $x[t]$ is not. With a single valid sample $J = e_0 e_0^T$ and $\eta = d[t]e_0$, giving $\hat{f} = d[t]$, $r = 0$, $\lambda = 1$: exactly right, since one datum under a diffuse prior determines its own baseline. In float32 that exact answer survives at $k = 2$, where rank deficiency forces $\text{range}(J) = \text{span}(e_0)$ and hence a *diagonal* J , so the factorization is trivial and $r = 0$ comes out bitwise. It is not relied on: (2.12) cuts these samples anyway.

2.5 Response and numerics

Transfer function. Where the mask is constant over a long stretch, J_f reaches the fixed point of its Riccati recursion and the estimator degenerates to a fixed IIR filter. For the two-sided limit $L \rightarrow \infty$, (2.6) diagonalizes in Fourier space and

$$H(\omega) = \frac{1}{1 + \rho (2 \sin(\omega/2))^{2k}}, \quad 1 - H(\omega) = \frac{\rho (2 \sin(\omega/2))^{2k}}{1 + \rho (2 \sin(\omega/2))^{2k}}, \quad (2.15)$$

monotone in $|\omega|$ with no stopband ripple, against the 24% ripple that §1’s truncated polynomial kernel has near $\omega \approx 2.2\pi/W$. Monotonicity is immediate here, since $\sin(\omega/2)$ is monotone on $[0, \pi]$ and $z \mapsto z/(1+z)$ is increasing. This is the main statistical argument for the estimator. The kernel at zero lag is

$$h[0] = \frac{c_k}{\tau} = \lambda_{\text{full}}, \quad (2.16)$$

so a pulse of width $w \ll \tau$ loses a fraction $\approx c_k w/\tau$ of its flux into the trend, the detrender removes $\approx c_k/\tau$ of the variance, and the requirement is $\tau \gg W_{\text{max}}$, the widest searched boxcar. (2.16) is also a closed-form check on the whole construction, and a strong one, since it exercises the forward recursion, the backward recursion and the combine (2.9) together against an analytic answer rather than against another implementation: run (2.10) and (2.11) to steady state on a full mask with $L \gg \ell$, and $h[0]$ – read off the equivalent kernel by impulses, so that nothing beyond the shipped outputs is needed – must approach c_k/τ . It does, from above and as $O(\tau^{-2})$: measured relative deviations $3.0 \times 10^{-2}, 7.7 \times 10^{-3}, 2.0 \times 10^{-3}$ at $\tau = 2, 4, 8$. This is in the test suite (§2.8). Both the history behind the output sample and the lag ahead of it must be several ℓ , or the forward filter is still spinning up and that dominates the deviation instead. Finite L perturbs all of this by $O(e^{-L/\ell})$; note that truncating the future also breaks the kernel’s symmetry, so H_L has a small phase response where the two-sided smoother and the local polynomial fit have none.

Conditioning. $\text{cond}(J_f)$ grows as $\tau^{2(k-1)}$, because the state components $\Delta^j f$ carry scales differing by $\tau^j - 1.3 \times 10^5$ at $k = 2$, $\tau = 256$. That is a pure diagonal scaling, under which Cholesky is equivariant (§1.5), and the quantity that matters is the equilibrated condition number $J_{ij}/\sqrt{J_{ii}J_{jj}}$, which is *flat* in τ : measured 5.83 at $k = 2$ across $\tau = 4, 16, 64, 256$, against 23 for the local polynomial fit’s Gram matrix. So no rescaling of the state is needed or used, and A, A^{-1} stay exact integer matrices. Two regimes do not spoil this. The matrix actually solved is better conditioned than J_f alone, since combining forward and backward information nearly symmetrizes it: at $k = 2$ its equilibrated condition number tends to 1 as τ grows (1.016 at $\tau = 64$, 1.004 at $\tau = 256$), the steady-state pivot ratios are (1, 1), and $r_{\min} \rightarrow 1$ – three decades above ϵ . And iterating J_f through a long masked gap with no measurements, the raw condition number runs away but the equilibrated one *plateaus* by $G \approx 10\tau$ at 13.9, with $J_{00} \sim 1/(qG^{2k-1})$ decaying only polynomially, so it cannot underflow for any realistic gap. (Inside such a gap (2.6) reduces to $\Delta^{2k}\hat{f} = 0$, so the estimate is the degree- $(2k-1)$ discrete spline interpolating between the surrounding data – it degrades gracefully rather than blowing up, though its variance does grow across the gap, which is unavoidable and physical.)

Float32. Everything runs in float32. Two things are needed and both are cheap. The constant offset of §2.7, without which the residual is a cancelling difference of two $|d|$ -sized numbers; and ϵ clearing the float32 noise in $r_{\min}$, since the mask expansion is the only place float32 feeds back into a discrete decision. That noise is measured at 3×10^{-7} , so $\epsilon = 10^{-3}$ clears it by a factor of 3500. Measured worst-case float32-against-float64 residual over the randomized-mask suite is $3 \times 10^{-7}\sigma$, and – the property the carried state depends on – it does *not* grow along a stream: the forward recursion is exponentially forgetting, so roundoff injected at time u contributes $O(e^{-(t-u)/\ell})$ and the error is bounded rather than accumulating.

2.6 Parameters

Order. $k = 2$. The naive reading points the other way – the *guaranteed* reproduction degrees are $k - 1 = 1$ here against $n = 2$ in §1 – but the guaranteed degree is not what governs the passband; the asymptotic one is, and there $k = 2$ already gives $2k - 1 = 3$, matching the local fit’s $n + 1 = 3$. So $k = 3$ would buy $O(\omega^6)$ rolloff, more than the other detrender has. It would also cost: the equilibrated condition number rises from 5.83 to 35.5, and the degenerate window of §2.4 stops being benign, since for $k \geq 3$ the deficient J is no longer diagonal and a floored pivot leaks into $\hat{x}_0$.

Timescale and lag. Take (k, τ, L) as the parameters, with τ and L in samples; ρ and q are machine quantities with enormous dynamic range and change meaning with k . To match §1 at half-width W , equate

the kernels at zero lag, $9/8W$ there against c_k/τ here:

$$\tau = \frac{8c_k}{9} W = 0.314 W \quad (k=2), \quad (2.17)$$

which also matches the flux loss and the degrees of freedom removed. L then follows from ℓ : the influence of data beyond $t+L$ decays as $e^{-L/\ell}$, and $L \approx 4\ell$ leaves a $\sim 1\%$ deviation from the ideal two-sided estimate. At the local fit's $W = 256$ that gives $\tau = 80.5$, $\ell = 114$, $L \approx 455$.

The lookahead is the real cost of this detrender: $L \approx 4\ell = 1.8W$, against exactly W for §1, so matched on smoothing strength it roughly doubles the latency. Against that, it needs no prepadding at all.

2.7 Implementation and cost

Chunking and state. On the i -th call the buffer is chunk i plus L samples of postpadding, so $T_c + L$ samples, and the carried state is (J_f, η_f) plus the constant offset $\kappa - O(k^2)$ per row, five floats at $k=2$. Outputs are committed once and never revisited.

The constant offset. Intensity data is total power, so d carries a large positive mean while everything of interest sits at the σ level on top of it. Subtract κ , the masked mean of the buffer in hand, before filtering. This is mathematically inert – the basis reproduces constants, since $\deg 0 \leq k-1$, so it shifts $\hat{f}$ by exactly κ and leaves the residual unchanged – and exists only so that the filter carries numbers of size $|d - \kappa|$. Unlike §1.5, κ cannot simply be recomputed per buffer and forgotten, because the state is expressed relative to it. Rebasings is exact and needs no special case: shifting the data by $\Delta = \kappa' - \kappa$ shifts the level component of the state, so $\mu \rightarrow \mu - \Delta e_0$ with J untouched, i.e.

$$\eta_f \longrightarrow \eta_f - \Delta J_f e_0. \quad (2.18)$$

At stream start $J_f = \eta_f = 0$ and this is a no-op. If a buffer has no valid samples, keep the previous κ rather than collapsing to zero, which would silently disable the subtraction.

Masked samples must be selected, never weighted. A masked sample may hold anything at all, including NaN, and $0 \times \infty$ is NaN. This matters more here than in §1: a NaN reaching the carried state destroys every subsequent output of that row, where there it would be confined to one buffer.

Cost. The forward pass is $O(k^2)$ per sample and is computed once. The backward factor is the expensive part. Evaluating (2.11) independently for each output is $O(Lk^3)$ per sample – $\sim 10^4$ flops at $L = 512$, $k = 2$ – which is fine for a reference implementation but not for a kernel. It is not the only option: the composition of these Gaussian potentials is associative, so the same van Herk block decomposition as §1.4 applies, with blocks of length $B = L$ so that $[t+1, t+L]$ spans exactly two adjacent blocks, prefix and suffix scans within a block, and one merge per output. That brings the cost to $O(k^3)$ per sample, of order 600 flops at $k = 2$, comparable to the ≈ 400 of §1.6. The monoid element is larger than the moment monoid's $3n+2$: for an interval it is the Gaussian potential relating the state at one end to the state at the other, $2k^2 + 3k = 14$ numbers at $k = 2$. Note also that a sequential sweep is numerically acceptable here, unlike in §1.4: the recursion forgets exponentially, so roundoff injected at one step decays rather than persisting, and tree depth is not load-bearing – only parallelism is.

2.8 Validation

Implemented in `pirate_frb/detrending/kf1d` and run by `python -m pirate_frb test --dtk1`. The checks worth keeping as regressions:

- **Against a dense oracle:** for each output, assemble (2.6) on $[0, t+L]$ explicitly and solve it with dense linear algebra, sharing no code with the recursions. Comparing the forward filter alone and the combination (2.9) pins both recursions, since $J_b = J - J_f$ is then determined by difference. This is the

test that fixes the signs and the A versus A^{-T} placement, and nothing else should be trusted before it passes.

- **Polynomial reproduction:** degree $\leq k - 1$ identically zero under arbitrary masks, at both precisions; and degree $2k - 1$ asserted from both sides – $O(1)$ near the stream start, suppressed as $e^{-L/\ell}$ in the interior – so that neither half can silently regress.
- **Chunk invariance:** several chunk sizes against a whole-stream call, requiring bit-identity for m_{out} and r_{min} unconditionally and for the residual at fixed κ . Writing this as a single tolerance over all three outputs would discard the unconditional half, which is the half worth having.
- **Steady-state response:** on a full mask, the equivalent kernel at zero lag must approach $h[0] = c_k/\tau$ as in (2.16), from above and as $O(\tau^{-2})$. Like the local fit’s zero-lag check this is analytic, so it needs neither the dense oracle nor a statistical tolerance, and it exercises both recursions and the combine at once. Note the projection identity of §1.7 has no analogue here: $\sum_u m[u]H_{tu}^2 = \nu \neq \lambda$, since a penalized estimator shrinks rather than projects.
- **Positive semidefiniteness and finiteness** over the randomized masks, including all-masked rows and rows with a single valid sample, with $\beta \geq 1/q$ checked at every step. The tolerance has to be per-precision, or float32 roundoff hides a genuine float64 failure.
- **Masked data unread:** poison masked samples with NaN/Inf and require every output *and the carried state* to be bit-identical. The state half is what keeps the hazard above from being live.
- **Float32 versus float64**, which is a test of understanding rather than a debugging test: it asks whether the error is where §2.5 says it is. It should report the fp32 noise in r_{min} , which is what ϵ must clear, and check that the discrepancy does not grow along a long stream.

3 2-d detrending

The detrenders of §1 and §2 work one frequency channel at a time, so by construction they cannot see structure *along* the frequency axis: bandpass ripple, coarse channel edges, gain steps at the boundary of an RFI-flagged region. This section describes a third detrender which fits both axes at once – a B-spline in frequency whose coefficients are allowed to drift as a low-order polynomial in time – and subtracts it. It is implemented in `pirate_frb/detrending/lps2d` and run by `python -m pirate_frb test --dt12`.

It is meant to run *after* one of the 1-d time detrenders and before the tree gridding kernel (*dedispersion.pdf*, “Tree gridding kernel”). The ordering is deliberate: this detrender is *regularized*, so unlike §1 it carries a small shrinkage bias, and the bias is proportional to the roughness of the baseline it is handed. Removing the large per-channel time baselines first is therefore the cheapest way to reduce it (§3.8).

Everything structural in this section follows from one difference: where §1 *masks* the samples it cannot fit, this one *regularizes* them. A masked B-spline fit has no analogue of the moving window’s “ $N_v \leq n$, drop it” rule – a coefficient whose support happens to be entirely flagged leaves an exactly zero row in the normal equations, and cleaning up the resulting contamination costs a large fraction of the surviving channels. Adding a small penalty makes the system positive definite instead, and converts a structural failure into a bounded bias. The rest of the design is about what that penalty must be, what it costs, and how the residual ill-conditioning is detected.

Some scientific context: empirically in CHIME, we find that a 2-d detrender seems to be needed, in order to remove *sudden*, *broadband* changes in total power prior to dedispersion. If we omit the 2-d detrender, then we find many false-positive FRBs. This is unfortunate, since the 2-d detrender produces a significant SNR penalty for some FRBs (the SNR penalty is a decreasing function of the DM-to-width ratio).

We don’t understand the underlying reason why these sudden, broadband changes in total power arise (or even whether they’re instrumental, or related to the RFI environment). It’s my hope that in the future, we can find a better way of dealing with them, and phase out the 2-d detrender. In the meantime, I don’t want to “design out” the option of using the 2-d detrender, so it’s implemented throughout the `pirate` software.

One consequence of this picture: it’s hard to predict in advance what 2-d detrender parameters (N_ϕ , K , n etc, see below) we’ll want to use. The choice of parameters is empirical, and may change in the future (if we deploy new hardware, or if the RFI environment changes). Therefore, we implement a flexible 2-d detrender, with a lot of parameters that can be varied.

3.1 Problem statement

The input is a float array $d[f, t]$ and a boolean mask $m[f, t]$ over $0 \leq f < N_{\text{freq}}$, with a leading spectator axis over beams which is not written below. As in §1.1 a sample is *valid* if $m = 1$ and junk otherwise, and we model the valid samples as

$$d[f, t] = F[f, t] + \epsilon[f, t], \quad (3.1)$$

with ϵ white and F a baseline which is smooth in f – meaning representable by the spline basis below – and slowly varying in t .

The window, and why it is short. Fix an output time t and a half-width W ; the fit uses the $2W + 1$ time samples $t + s$, $s \in \{-W, \dots, W\}$, with the spline coefficients modelled as polynomials of degree n in s and evaluated back at $s = 0$. This is the same local polynomial subtraction as §1.2, applied to the coefficients rather than to the data. But W here is a handful of samples, not the hundreds of §1.1, because it is doing a different job. The frequency fit is what removes the baseline; the window exists to keep the fit from overfitting. At $W = 0$ the fit spends N_ϕ degrees of freedom on N_{freq} channels every sample, and the leverage is $\sim N_\phi/N_{\text{freq}}$; the window divides that by $(2W + 1)/(n + 1)$. Its cost is a bias if the baseline drifts faster than degree n across $2W + 1$ samples, so W is bounded above by the drift timescale and below by the leverage one is willing to accept. $(n, W) = (0, 0)$ – an independent frequency fit per time sample – is a corner of the same code path rather than a separate one.

Outputs. Three arrays, matching §1.1 except in shape. The residual $r[f, t] = d[f, t] - \hat{F}[f, t]$; an expanded mask $m_{\text{out}}[f, t]$, zero where the residual is meaningless; and the conditioning statistic $r_{\text{min}}[t, z]$ of (3.16), which is *per zone* rather than per channel. A zone (§3.2 below) is a contiguous block of channels whose fit is exactly decoupled from the rest of the band, and it is the unit in which everything here is decided: the fit is independent per zone, the conditioning statistic is per zone, and mask expansion drops a whole zone at a time. That is a real difference from §1, where each output sample is masked on its own merits: here the channels of one zone at one time either all survive or all do not, because they share a factorization.

The detrender runs incrementally, and **the caller owns the padding**. The i -th call is handed $T_c + 2W$ time samples – chunk i plus W of pre- and postpadding – and emits exactly T_c . Nothing is emitted in the padding regions and no state is carried across calls, so each call is a pure function of its arguments: there is no stream-boundary policy, no first-chunk special case, chunks may be processed in any order, and replay is bit-reproducible. Supplying padding that is consistent across chunks is the caller’s responsibility.

symbol	meaning	value or range
n_ϕ	spline degree in frequency	$0 \dots 3$; 2 typical
N_ϕ	number of B-splines	$K + n_\phi$ if interior knots are simple
K	number of non-empty knot intervals	≤ 10 tested
h	widest knot interval, in channels	up to N_{freq}
N_z	number of zones	≥ 1
n	degree of the time polynomial	0, 1 or 2
W	window half-width, $2W+1 \geq n+1$ required	a few samples
η	regularization strength	10^{-3}
ϵ	mask-expansion threshold on r_{min}	3×10^{-5} (fp32), 10^{-7} (fp64)
	arithmetic	float32

No production (n_ϕ, n, W, K) has been frozen; the code is written for general values and the test suite sweeps $n_\phi \in \{0, 1, 2, 3\}$, $(n, W) \in \{(0, 0), (0, 1), (0, 3), (1, 1), (1, 2), (2, 1), (2, 2), (2, 4)\}$ and N_{freq} up to 3×10^4 . η and ϵ are *not* free: they are tied to each other by a measured scaling law (§3.7).

3.2 The B-spline basis

A brief review, in the form the detrender uses. Fix a degree $n_\phi \geq 0$ and a non-decreasing integer *knot vector* $k_0 \leq k_1 \leq \dots \leq k_{N_\phi+n_\phi}$ with values in $[0, N_{\text{freq}}]$. The N_ϕ basis functions are given by the Cox-de Boor recursion, starting from the indicator of a knot interval and raising the degree one step at a time:

$$\phi_j^{(0)}(x) = \begin{cases} 1 & k_j \leq x < k_{j+1} \\ 0 & \text{otherwise,} \end{cases} \quad \phi_j^{(r)}(x) = \frac{x - k_j}{k_{j+r} - k_j} \phi_j^{(r-1)}(x) + \frac{k_{j+r+1} - x}{k_{j+r+1} - k_{j+1}} \phi_{j+1}^{(r-1)}(x), \quad (3.2)$$

with the convention $0/0 \equiv 0$, and $\phi_j \equiv \phi_j^{(n_\phi)}$. Repeated knots are fully supported and are used heavily below, which is why the implementation evaluates (3.2) in its triangular form (the NURBS book’s Algorithm A2.2) restricted to the $n_\phi + 1$ functions that are nonzero at x : in that form no denominator can vanish at all, provided the evaluation point’s own knot interval is non-empty.

Four properties are used, and only these four:

- **Compact support.** $\text{supp } \phi_j = [k_j, k_{j+n_\phi+1})$, so at most $n_\phi + 1$ basis functions are nonzero at any x . This is what makes the Gram matrix banded with half-bandwidth n_ϕ , and what makes the per-channel work $O(n_\phi)$ rather than $O(N_\phi)$.
- **Nonnegativity.** $\phi_j \geq 0$. Used in the bound (3.15).
- **Smoothness by multiplicity.** At a knot of multiplicity μ the spline is $C^{n_\phi-\mu}$. Multiplicity is thus a design knob and not a degenerate case: it is how a known discontinuity in the bandpass is put into the model rather than smoothed across.

- **Partition of unity.** $\sum_j \phi_j(x) = 1$ wherever the basis is complete. This is the one the whole design rests on, because it says the all-ones coefficient vector *is* the constant function.

Conventions. Channel f occupies $[f, f+1)$ and its datum sits at $f + \frac{1}{2}$, so knot values are channel indices, no data point ever coincides with a knot, $k_{i+1} - k_i$ is literally a channel count, and the interval test is the ordinary half-open integer one. The first and last knots have multiplicity exactly $n_\phi + 1$ (*clamped ends*), which is what makes the basis complete – hence a partition of unity – on the whole band $[0, N_{\text{freq}}]$ rather than on the interior knot span only. This is not a stylistic choice and it does not degrade gracefully: at end multiplicity n_ϕ the best fit to the constant function 1 is already off by 0.99 near the band edge.

Zones. An interior knot of multiplicity $n_\phi + 1$ is a *zone boundary*. No basis function straddles one: if the boundary occupies knot indices $i \dots i + n_\phi$ with common value v , then $j \leq i - 1$ gives $\text{supp } \phi_j \subset [\cdot, k_{i+n_\phi}) = [\cdot, v)$ and $j \geq i$ gives $\text{supp } \phi_j \subset [v, \cdot)$. Consequently

$$G \text{ and } D_1 \text{ are \textit{exactly} block diagonal across a zone boundary,} \quad (3.3)$$

the fits on the two sides are independent, the partition of unity holds *within each zone*, and the fitted baseline is allowed to jump discontinuously at the boundary. All three matter: the first is why flagging one zone can never affect another, the second is the null-space statement of §3.4, and the third is how a hard gain step is represented exactly instead of being smeared over $n_\phi + 1$ knot intervals.

3.3 The estimator

Let $\psi_0, \dots, \psi_n$ be a basis for the polynomials of degree $\leq n$ on the window offsets $s \in \{-W, \dots, W\}$ (the specific choice is fixed in §3.6, and is orthonormal). The model over the window is the tensor product

$$b(f, s) = \sum_{j=0}^{N_\phi-1} \sum_{q=0}^n \alpha_{jq} \phi_j(f) \psi_q(s) = \sum_j a_j(s) \phi_j(f), \quad a_j(s) = \sum_q \alpha_{jq} \psi_q(s), \quad (3.4)$$

so that $b = \Phi \alpha$, where Φ is the design matrix with row index (f, s) , column index (j, q) and entries $\phi_j(f) \psi_q(s)$. α minimizes

$$\chi^2(\alpha) = \sum_{s=-W}^W \sum_f m[f, t+s] \left(d[f, t+s] - b(f, s) \right)^2 + \eta \sum_{s=-W}^W \sum_{(j,j+1)} \left(a_{j+1}(s) - a_j(s) \right)^2, \quad (3.5)$$

the second sum running over pairs of adjacent coefficients *within a zone*. The committed baseline is the fit evaluated back at the window centre,

$$\hat{F}[f, t] = \sum_j a_j(0) \phi_j(f), \quad a_j(0) = \sum_q \alpha_{jq} \psi_q(0), \quad (3.6)$$

which is α_{j0} only for a monomial basis; with an orthogonal one every even q contributes.

Sufficient statistics. Define, at each individual time sample u ,

$$G_{jl}[u] = \sum_f m[f, u] \phi_j(f) \phi_l(f), \quad U_j[u] = \sum_f m[f, u] \phi_j(f) d[f, u]. \quad (3.7)$$

These are sufficient for everything the time axis does, and they are the entire interface between the two axes: $N_\phi(n_\phi + 2)$ floats per (beam, time) against N_{freq} , a compression of about 60 at $N_{\text{freq}} = 4096$, $K = 16$, $n_\phi = 2$. The full-resolution arrays are therefore touched exactly twice – once to form (G, U) , once to evaluate (3.6) and subtract – and everything in between runs on the reduced arrays.

Minimizing (3.5) gives the normal equations $\mathcal{G}\alpha = \mathcal{U}$ in the composite index $I = (j, q)$:

$$\boxed{\mathcal{G}_{(jq),(lr)} = \underbrace{\sum_{s=-W}^W \psi_q(s)\psi_r(s) G_{jl}[t+s]}_{\mathcal{M}_{jl}^{qr}} + \eta (D_1)_{jl} \Theta_{qr}, \quad \mathcal{U}_{(jq)} = \sum_{s=-W}^W \psi_q(s) U_j[t+s],} \quad (3.8)$$

where D_1 is the first-difference matrix of (3.5) and

$$\Theta_{qr} = \sum_{s=-W}^W \psi_q(s)\psi_r(s) \quad (3.9)$$

is the *window* Gram of the time basis. In tensor notation the regulator is $\eta \mathcal{R}$ with $\mathcal{R} = D_1 \otimes \Theta$.

Four things about (3.8) are worth stating explicitly.

The regulator never sees the mask. Θ is a sum over the full window and D_1 depends only on the knot vector, so $\mathcal{R}$ is a fixed matrix, identical at every (f, t) . That is deliberate, and it is what makes the bias predictable rather than mask-dependent; it is also what makes the exact worst-case mask construction of §3.10 a linear-fractional program rather than something intractable.

Accumulate, then solve – not solve, then smooth. $\mathcal{M}$ sums G over the window *before* inverting. The distinction is invisible when nothing is masked and matters when something is, because $\bar{G}^{-1}\bar{U}$ is not the average of the $G[t+s]^{-1}U[t+s]$, and only the former is the least-squares estimator of (3.5). It also behaves better: a time sample whose own G is rank deficient still contributes the information it has, instead of producing a garbage coefficient that is then averaged in.

A stencil, not a van Herk decomposition. $\mathcal{M}^{qr}$ is a fixed-coefficient correlation of G along t , evaluated directly, at $O(W)$ per output per moment. The monoid and block scan of §1.3–§1.4 are $O(1)$ in W and are deliberately not ported: at $W \sim 1 \dots 4$ they cost the same to within a flop per full-resolution sample, and the stencil buys no block lattice, no prefix/suffix storage, no carried partials, parallelism in t , and the purity of the chunk contract. Because the window is symmetric and the ψ_q have definite parity, each stencil is either even or odd in s and the window folds in half,

$$\text{even: } \sum_{k>0} c_k (G[t+k] + G[t-k]) + c_0 G[t], \quad \text{odd: } \sum_{k>0} c_k (G[t+k] - G[t-k]), \quad (3.10)$$

which halves the multiplies and, more importantly, makes the odd moments vanish *exactly* rather than by cancellation of rounding errors. The exactness is load-bearing: “ $n = 1$ reduces to $n = 0$ whenever the mask is constant over the window” depends on it.

Banded, in one index order only. With $I = j(n+1) + q$ – coefficient major – we have $|I - I'| = |j - l|(n+1) + |q - r|$, and $\mathcal{M}$ vanishes for $|j - l| > n_\phi$, so $\mathcal{G}$ is banded with half-bandwidth

$$n_b = \max(n_\phi, 1)(n+1) + n. \quad (3.11)$$

The max is not cosmetic: the two contributions to (3.8) have *different* bandwidths in j , n_ϕ for the data block and 1 for the regulator whatever n_ϕ is, since D_1 is a difference penalty on coefficient indices and knows nothing about the spline degree. With the other natural order, $I = qN_\phi + j$, the half-bandwidth is $nN_\phi + n_\phi$, which grows with N_ϕ and is effectively dense (measured 2/5/8 coefficient-major at $n = 0/1/2$ against 2/19/36 time-major, at $n_\phi = 2$, $N_\phi = 17$). Since the solve is banded, the wrong order would silently turn an $O(Nn_b^2)$ factorization into an $O(N^3)$ one. Zones stay contiguous under this ordering because j is the major index, so (3.3) survives it.

3.4 The regulator

The penalty in (3.5) is a first difference on *coefficient indices* – a P-spline penalty in the sense of Eilers and Marx – assembled per zone. Three properties recommend it over the textbook $\int (b')^2 df$:

- Same null space (the constants) and, measured, the same smallest nonzero eigenvalue to 2%, hence the same bias at matched regularization strength.
- Half-bandwidth 1 in j instead of n_ϕ .
- It never differentiates anything, so repeated knots cannot break it. $\int (b^{(m)})^2$ *diverges* at a knot of multiplicity $\mu > n_\phi - m + 1$, and – the real hazard – per-interval quadrature returns a finite matrix anyway, with a spurious extra null direction that silently destroys the positive-definiteness guarantee below. Since repeated interior knots are exactly how zone boundaries and reduced-continuity points are expressed (§3.2), this is not an exotic corner.

Assembly per zone is likewise not optional: a difference penalty built across a zone boundary would couple the two sides with weight 1, destroying both the block structure (3.3) and the exactness statement below. Since the coefficients of a zone are contiguous, “per zone” just means dropping the differences that straddle a boundary.

What is reproduced exactly. Θ is positive definite, so $\alpha \in \ker \mathcal{R}$ iff α_{jq} is independent of j within each zone for every q ; and by the per-zone partition of unity (§3.2) the all-ones coefficient vector of a zone is the constant function. Hence

$$\ker \mathcal{R} = \left\{ b(f, s) = c_z P(s) : c_z \text{ constant on each zone, } \deg P \leq n \right\}, \quad (3.12)$$

and any baseline of that form – constant in frequency within each zone, times an arbitrary degree- n polynomial in time – is removed **exactly, at any η , under any mask**. This is the analogue of the polynomial reproduction of §1.7, and like it, the expected answer is analytic, which makes it the sharpest test available.

When the system is positive definite. This is the statement that replaces the “ $N_v \leq n$ ” rank test of §1.2, and it is a rank condition on the *window*, not a channel count.

Proposition. Zone z ’s block of $\mathcal{G}$ is positive definite if and only if the zone holds at least one valid channel at each of $\geq n + 1$ *distinct* offsets s in the window.

Proof. The data block is positive semidefinite, so a null vector of $\mathcal{G}$ must lie in $\ker \mathcal{R}$, i.e. be of the form (3.12): $b(f, s) = P(s)$ on the zone, with $\deg P \leq n$. For such a vector

$$\alpha^T \mathcal{G} \alpha = \sum_{s=-W}^W L_z(s) P(s)^2, \quad L_z(s) = \#\{f \in z : m[f, t + s] = 1\}, \quad (3.13)$$

which vanishes iff P vanishes at every offset carrying data. A nonzero polynomial of degree $\leq n$ has at most n roots, so the sum is strictly positive exactly when $n + 1$ offsets carry data.

So one valid channel at each of $n + 1$ offsets suffices, while thousands of valid channels all at a single offset do not. At $(n, W) = (0, 0)$ the condition reads “the zone holds one valid channel”. It is evaluated structurally, by counting the offsets at which the zone’s diagonal of G is nonzero, rather than inferred from the size of a pivot: the condition is exact and combinatorial, and a magnitude test would only approximate it. A zone failing it reports $r_{\min} = 0$ and is masked.

The bias, which is the price. For noiseless data lying exactly in the span, $\mathcal{U}$ is the data block applied to the true α , so

$$\hat{\alpha} = (I - \eta \mathcal{G}^{-1} \mathcal{R}) \alpha, \quad \delta \alpha \equiv \alpha - \hat{\alpha} = \eta \mathcal{G}^{-1} \mathcal{R} \alpha, \quad (3.14)$$

and the leftover baseline is (3.6) applied to $\delta \alpha$. Two consequences. The bias is proportional to $\mathcal{R} \alpha$, i.e. to the baseline’s *roughness on the knot scale* and not to its amplitude – which is (3.12) again, read quantitatively.

And its size is $O(\eta)$ times the amplitude: measured over the randomized-mask suite at $\eta = 10^{-3}$, the median is 5.4×10^{-4} of the baseline amplitude and the worst seen is 6×10^{-3} . $I - \eta \mathcal{G}^{-1} \mathcal{R}$ is the *shrinkage operator*, with spectrum in $[0, 1]$; it appears again in §3.8.

3.5 Conditioning: equilibration, $r_{\min}$, and mask expansion

The solve is: equilibrate $\mathcal{G}$ to unit diagonal, factor by a banded Cholesky recording every pivot, solve, unscale. The order is the specification, not an implementation choice.

Equilibration is load-bearing twice over. As in §1.5 it is what makes a threshold on a pivot scale-invariant, so that ϵ needs no calibration against the units of the data. Less obviously – and this is the more important half here – it is what makes the problem well conditioned at all. The raw matrix $\mathcal{G}$ has condition number $O(h/\eta)$: a coefficient with no data under its support has diagonal $\eta(D_1)_{jj}\Theta_{qq}$, while one sitting in the middle of a wide populated knot interval has diagonal $O(h)$. At $h = 3000$, $\eta = 10^{-3}$ that is already $\sim 10^5$, and it grows without bound as η falls. Dividing row and column by $\sqrt{\mathcal{G}_{II}}$ cancels *both* factors exactly – the η on the dead coefficients and the h on the live ones – and the equilibrated condition number is $O(1)$ in both. This is the standard van der Sluis result again, but here it is the difference between a tractable problem and an intractable one rather than a factor of a few.

The upper end is then pinned analytically. Writing $\hat{\mathcal{G}}$ for the equilibrated matrix,

$$\lambda_{\max}(\hat{\mathcal{G}}) \leq \max(2, (n_\phi + 1)(n + 1)). \quad (3.15)$$

Proof sketch. For a rank-one xx^T with ν nonzero entries, $\nu \text{diag}(xx^T) - xx^T \succeq 0$. The data block is a nonnegative combination of such terms, one per valid (f, s) , each with $\nu = (n_\phi + 1)(n + 1)$ nonzeros by compact support. For the regulator (with $\Theta = I$, §3.6), $2 \text{diag}(D_1) - D_1$ is the signless Laplacian $\sum_{\text{edges}} (e_j + e_{j+1})(e_j + e_{j+1})^T \succeq 0$. Hence $\max(2, (n_\phi + 1)(n + 1)) \text{diag}(\mathcal{G}) - \mathcal{G} \succeq 0$. Checked numerically over random $(n_\phi, n, W, \text{knots}, \text{mask})$: the bound holds in every case and is attained. So the whole conditioning question is a *lower* bound on $\lambda_{\min}(\hat{\mathcal{G}})$.

The statistic. Let p_I be the Cholesky pivots of $\hat{\mathcal{G}}$ and define, per zone,

$$r_{\min}[t, z] = \min_{I \in z} p_I \in [0, 1], \quad r_{\min} \equiv 0 \text{ if the zone fails the rank test of §3.4.} \quad (3.16)$$

Since $\hat{\mathcal{G}}_{II} = 1$ this is exactly the smallest *relative* pivot of (1.9), the same statistic the 1-d detrender thresholds on, read off a bigger matrix. **The whole zone is masked when** $r_{\min} < \epsilon$, and that is the only expansion rule there is. With $\eta > 0$ there are no dead coefficients to deflate and no contamination to chase through the connected components of G 's zero pattern, which an unregularized spline fit would require; and by (3.3) flagging one zone can never affect another.

Why $r_{\min}$ rather than $\lambda_{\min}$. They bound different errors. Cholesky is backward stable, so the error in the *coefficients* is governed by $\lambda_{\max}/\lambda_{\min}$, and by (3.15) that is $O(1/\lambda_{\min})$. $r_{\min}$ is not that quantity: pivots are diagonal entries of Schur complements, so $\lambda_{\min} \leq r_{\min} \leq 1$ always, with a gap reaching 84× on real detrender matrices. But the coefficients are not what is emitted. The residual is evaluated at *valid channels*, and for that error we have no rigorous bound in terms of either statistic – only a measurement, over 523 random and adversarial configurations spanning four decades of $r_{\min}$, in float32:

$r_{\min}$	worst fit error
$[10^{-1}, 1)$	1.9×10^{-7}
$[10^{-2}, 10^{-1})$	1.7×10^{-6}
$[10^{-3}, 10^{-2})$	1.9×10^{-5}
below 10^{-3}	1.2×10^{-4}

i.e. roughly $\epsilon_{\text{mach}}/(4r_{\min})$. *This is an empirical rule of thumb and not a bound:* the log-log slope against $1/r_{\min}$ is 0.68 rather than 1 and the correlation is 0.62, so “no case exceeded $\epsilon_{\text{mach}}/r_{\min}$ ” is an observation

about the masks we generated rather than a theorem. A rigorous statement would need the componentwise (Skeel) bound $|\mathcal{G}^{-1}||\mathcal{G}||\alpha|$, which is not cheap to evaluate.

With that caveat $r_{\min}$ is still the right choice of the two, for a structural reason rather than a lucky one. The small modes of $\lambda_{\min}$ are typically *delocalized over coefficients with no data* – the extremal one is a ramp across a dead run of K coefficients, giving $\lambda_{\min} \sim 1/K^2$ – and such a mode contributes nothing at a valid channel. $r_{\min}$ is a local statistic and does not see it. Thresholding on $\lambda_{\min}$ would therefore expand the mask on zones whose fits are perfectly accurate: measured at $N_{\text{freq}} = 10^4$ with no interior knots, $n_\phi = 2$ and $\eta = 3 \times 10^{-3}$, $\lambda_{\min} = 8.6 \times 10^{-5}$ against $r_{\min} = 5.1 \times 10^{-4}$, so the two thresholds would fire a factor of 6 apart on identical data.

3.6 The time basis, and why ϵ carries over unchanged

$\{\psi_q\}$ is taken **orthonormal** on the window – Gram-Schmidt (in practice QR) applied to the monomials s^q – so that $\Theta = I$ in (3.9). This is the one choice in the section that moves a threshold rather than a constant, and the reason is a small exact result.

Suppose the mask is constant across the window, $m[f, t + s]$ independent of s . Then $G[t + s] = G$ and (3.8) collapses to a Kronecker product,

$$\mathcal{G} = (G + \eta D_1) \otimes \Theta. \quad (3.17)$$

Equilibration of a Kronecker product is the Kronecker product of the equilibrations, and $L_{A \otimes \Theta} = L_A \otimes L_\Theta$, so the relative pivots *multiply*:

$$r_{\min}^{2d} = r_{\min}^{1d} \cdot r_{\min}(\Theta). \quad (3.18)$$

With raw monomials $r_{\min}(\Theta)$ is 0.18 to 0.25 at $n = 2$ (tending to $1 - \sqrt{5}/3$, worst at small W), so the entire 1-d conditioning margin would be multiplied by that factor – enough to push the worst adversarial mask below ϵ and expand zones whose fits are perfectly accurate. With $\Theta = I$, $r_{\min}^{2d} = r_{\min}^{1d}$ *exactly*, and the threshold ϵ therefore carries over from the pure frequency fit with no recalibration. It costs nothing: the basis enters only through stencil coefficients that are precomputed either way.

Two footnotes. Orthogonalizing does forfeit the Hankel structure that monomials have – with $\psi_q = s^q$ the block depends on (q, r) only through $q + r$, so there are $2n + 1$ distinct moment arrays rather than $(n + 1)(n + 2)/2$ – which is one extra stencil at $n = 2$ and nothing else. And (3.17) needs the mask constant *across the window*, which is strictly stronger than separable: for $m[f, t] = m_f m_t$ the data block is $G_f \otimes \Theta_{\text{valid}}$ but the regulator still carries the full-window Θ , so the sum is Kronecker only when $m_t \equiv 1$. The distinction is easy to miss and the identity fails immediately without it.

3.7 Choosing η and ϵ

η is dimensionless. D_1 is defined on coefficient indices, so it is already in index coordinates and needs no rescaling by the knot spacing; and both the shrinkage bias and the residual noise are W -independent, so one value serves every window. It wants to be *small*, because its cost is the undersubtraction of (3.14), which §3.8 argues is dangerous at any rate however rare.

What stops it is conditioning. The extremal behaviour is governed by the *widest knot interval* h , not by N_{freq} or by K , and follows a power law

$$\lambda_{\min}^{\text{worst}} \sim C \left(\frac{\eta}{h} \right)^{2r/(2r+1)}, \quad r = \min(n_\phi, 2), \quad (3.19)$$

i.e. exponent $2/3$ at $n_\phi = 1$ and $4/5$ at $n_\phi \geq 2$. The mechanism is that the extremal coefficient vector is one whose spline has a root of multiplicity r inside the wide interval, positioned so that the surviving data sees only the flat bottom of it; a triple root is available at $n_\phi = 3$ but does not pay for its D_1 energy, which is why r saturates at 2. Measured by exhaustive search over single contiguous valid runs at $K = 5$,

$\eta = 10^{-3}$, $h = 256 \dots 65536$, the ratio $\lambda_{\min}/(\eta/h)^{4/5}$ at $n_\phi = 2$ is 14.65, 14.82, 14.91, 14.96, 14.99 – flat to three digits – while the same numbers against $(\eta/h)^{2/3}$ drift by a factor of two. The optimal run length is $m_* \approx 0.06 \dots 0.10 h$. The worst-case $r_{\min}$ follows the same power of η : over five configurations from $N_{\text{freq}} = 10^4$ to 3×10^4 , cutting η by 3 multiplies $r_{\min}$ by $0.417 \dots 0.421$ against $3^{-4/5} = 0.415$ predicted.

So η and ϵ move together, and are chosen jointly. Worst-case $r_{\min}/\epsilon$ from an offline exhaustive sweep:

configuration	$(\eta, \epsilon) = (3 \times 10^{-3}, 10^{-4})$	$(10^{-3}, 3 \times 10^{-5})$
$N_{\text{freq}} = 10^4$, $K = 10$ uniform	23.3×	32.7×
$N_{\text{freq}} = 10^4$, no interior knots	5.1×	7.1×
$N_{\text{freq}} = 3 \times 10^4$, one 29900-wide interval	2.1×	3.0×
$N_{\text{freq}} = 3 \times 10^4$, no interior knots	2.1×	3.0×

Cutting η from 3×10^{-3} to 10^{-3} costs a factor 2.4 in $r_{\min}$ by (3.19), but lets ϵ come down by 3.3, so every margin *improves* by about 1.4 while the median shrinkage bias roughly halves.

Lowering ϵ is not free, and that is what sets the floor. A zone surviving at $r_{\min} \sim \epsilon$ carries a float32 fit error of order $\epsilon_{\text{mach}}/(4r_{\min})$, and that error is *coherent* – the same dangerous category as shrinkage, not the harmless one. Measured at each setting’s worst configuration it is 5.8×10^{-6} , 8.6×10^{-6} , 1.6×10^{-5} at $\eta = 3 \times 10^{-3}$, 10^{-3} , 3×10^{-4} , against shrinkage biases of 1.06×10^{-3} , 5.4×10^{-4} , 1.7×10^{-4} . At $\eta = 10^{-3}$ the numerical term is still $60\times$ below the regulator term; by 3×10^{-4} the two are within a factor of 2 of crossing. $\eta = 3 \times 10^{-4}$ is the floor – below it, one coherent error is simply traded for another.

ϵ is a threshold on a dimensionless pivot, so its natural scale is machine epsilon rather than anything physical; hence one value per working precision. The float64 value is nowhere near binding ($890\times$ margin at the worst configuration above) and is left where it is.

Finally, a fact about $r_{\min}$ that is worth internalizing because intuition points the wrong way: *a zone with a single valid channel is one of the safest masks there is*, $r_{\min} \approx 1.6 \times 10^{-2}$, some $40\times$ better than the worst. The null space contains the constants, so the fit simply interpolates that one point. The dangerous mask is a single contiguous run of a few hundred channels at a particular fractional offset inside one wide knot interval, with everything else flagged – a configuration nobody would write down by hand, which has consequences for testing (§3.10).

3.8 What is deliberately not computed

This subsection is the analogue of the leverage discussion in §1.1, and it turns on the same asymmetry, which is worth stating once in general form because it decides every threshold in this section. The search sweeps $\sim$ petabytes/day for ultra-rare $\sim 10\sigma$ outliers, so two failure modes are not comparable:

- **Undersubtraction** – shrinkage, or excess variance – leaves excursions in the data and converts noise into triggers. With large trial factors there are many 7σ noise fluctuations, so a mechanism that biases by 3σ with probability 10^{-5} can still flood the search and force the threshold up. *Dangerous at any rate, however rare.*
- **Overfitting** – leverage – subtracts noise instead of baseline and suppresses the residual. Converting a 20σ event to 10σ with probability 10^{-5} does not move the event rate. *Only a significant average effect matters.*

No residual-degrees-of-freedom cut. For a fixed mask the estimator is linear, $\hat{F} = Hd$ with $H = \Phi \mathcal{G}^{-1} \Phi^T \text{diag}(m)$ restricted to the valid channels, and the natural summary of overfitting is $\nu = M_z - \text{tr } H$ over a zone’s M_z valid channels. (Note $\text{tr}(\mathcal{G}^{-1}G) = \text{tr } H$ exactly, by cyclicity, so the two obvious ways of writing it are the same number.) It would be a real cut: a zone reduced to a single valid channel is fit *exactly*, has $\nu = 0$, and is nonetheless perfectly well conditioned, so $r_{\min}$ does not flag it. There is deliberately no ν cut, for three reasons.

1. Overfitting here has the *wrong sign* and cannot manufacture an excursion at all. Restricted to the valid samples H is symmetric positive semidefinite with spectrum in $[0, 1]$ – its nonzero eigenvalues are those of $I - \eta \mathcal{G}^{-1} \mathcal{R}$, the shrinkage operator of (3.14) – so $I - H$ and hence $(I - H)^2$ have spectrum in $[0, 1]$ too, and

$$\text{Var}[r] = \sigma^2 [(I - H)^2]_{ff} \leq \sigma^2 \quad (3.20)$$

identically. Measured maximum 0.9968 over 101 masks: over-dispersion never occurs. This is stronger than “small on average” – it is “wrong direction, always”.

2. Operationally the leverage is negligible and rigorously bounded: $\text{tr } H \leq N_\phi(n + 1)$, so leverage $\leq N_\phi(n + 1)/M_z$. Measured at $N_{\text{freq}} = 4096$: 0.0032 fully valid, 0.0035 at 10% flagged, 0.0064 at 50% flagged, each sitting on that bound.
3. It is not cheap. ν needs the banded entries of $\mathcal{G}^{-1}$ (a Takahashi recursion on the factor already in hand), per zone per time sample, for no false-positive protection.

One caveat is an interface concern rather than a detrander defect: at $\nu = 0$ the residual is identically *zero*, not merely small, so anything downstream that estimates a variance from the data and divides by it will meet 0/0 on such a zone. If that matters, the cheap fix is not ν but the count. $\text{tr } H \leq \min(M_z, N_\phi(n + 1))$, and $\sum_{j \in z} G_{jj}$ – already computed for the rank test of §3.4 – lies in $[M_z/(n_\phi + 1), M_z]$ by Cauchy-Schwarz and the partition of unity, so tightening its “ > 0 ” to “ $> c N_\phi(n + 1)$ ” is a complete, free and conservative surrogate.

No runtime undersubtraction statistic, though unlike the above this is deferred rather than settled. The bias (3.14) is the failure mode that lands in the dangerous category, so it deserves more than a scaling argument. The strongest lever is the pipeline order already described: the 1-d time detrander has no shrinkage, so running it first removes the large baselines before the regularized fit ever sees them, and since the bias is proportional to roughness this should reduce the whole problem proportionally. A runtime tripwire does exist in principle. From (3.8) there is an exact identity,

$$\eta \mathcal{R} \hat{\alpha} = \Phi^T \text{diag}(m) r, \quad (3.21)$$

which says the regulator term *is* the basis-projection of the residual – for an unregularized fit that projection is exactly zero by normal-equation orthogonality, so the departure from orthogonality is precisely the under-subtracted baseline. It costs one extra triangular solve on the factor in hand, and normalizing by $\sum_f m_f r_f^2$ gives a dimensionless, offset-independent ρ . What stops it from being a measurement is that substituting $\hat{\alpha}$ for α multiplies it by the shrinkage operator of (3.14), whose spectrum lies in $[0, 1]$, so it under-reports *always*, and worst exactly where shrinkage is worst (median estimate/true falls from 0.998 to 0.346 across four decades of true bias). It is monotone (log-log correlation 0.93), hence usable as a tripwire, and the blocking question is not the statistic but the threshold, which needs the operational distribution of ρ on real data after the 1-d stage.

No offset (κ) subtraction. Mathematically it would be inert: a global constant is a constant on each zone, hence in the span by the partition of unity, so the fit is translation-equivariant and the residual is unchanged. But it is exactly what protects float32 precision against a large DC level, for the reason spelled out in §1.5 – intensity data is total power, and every accumulator in (3.7) then sums terms of magnitude $|d|$ while the structure of interest lives at the σ level. Until it exists, feeding float32 data with a large offset relative to its structure loses mantissa bits for nothing.

3.9 Structure and cost

Only two passes touch the full-resolution arrays – $(d, m) \rightarrow (G, U)$ and coefficients $\rightarrow$ residual – and both are pure FMA against precomputed per-channel tables of ϕ_j and of the products $\phi_a \phi_b$ (9 floats per channel at $n_\phi = 2$). Everything else runs on the reduced arrays: the stencil (3.10) at $O(W)$ per output per moment, assembly, and a banded Cholesky of size $N = N_\phi(n + 1)$ and half-bandwidth (3.11), $O(N n_b^2)$.

Three points that are not implementation detail.

The reduction over frequency must be an explicit binary tree over fixed-size channel blocks, not a library sum. Accuracy is the lesser reason (a zone can span thousands of channels). The real one is reproducibility: numpy’s own pairwise summation blocks by *stride*, so its grouping over frequency changes silently with the length of the time axis, and the same output computed in a 1-sample chunk and a 24-sample chunk then differs in the last bits. A tree over a padded channel axis depends only on the channel count, so chunk invariance is exact. This is a property of the algorithm, being what makes “pure function of its arguments” (§3.1) true bit-for-bit rather than approximately.

The cost is independent of the mask. Both full-resolution passes sweep every channel with no early-out and no compaction. Same reasoning as §1.6: the real-time latency budget must be met in the worst case, and RFI is bursty, so a scheme that degrades under heavy flagging degrades for many channels at once.

There is no data-dependent control flow. Every mask-dependent decision – the accumulation, the equilibration guard, the pivot guards, the rank test, the residual – is a predicated select, not a branch, so a GPU port has no warp divergence to design around. That is not luck: the governing rule is *select masked samples away, never weight by the mask*, adopted because a flagged sample may hold anything including NaN from a dropped packet and $0 \times \text{NaN} = \text{NaN}$. The binding constraint for a kernel is instead the working set. Zones decouple, so the unit of work is a *zone* rather than the whole coefficient vector: the banded factor, right-hand side and equilibration scales are $11N_\phi^{\text{zone}}(n+1)$ floats, about 800 bytes in float32 at $n_\phi = n = 2$ with six basis functions per zone. That is past per-thread registers – and registers are unavailable in any case, since arbitrary runtime knots make the dimension a runtime quantity and the loops therefore unrollable – so one thread per (beam, zone, time sample) is workable provided the working set lives in *shared* memory, which is what `GpuDetrenderLps2d` does.

3.10 Validation

Two exact statements anchor the suite, and both are analytic, so neither needs a reference implementation or a statistical tolerance:

- **Exact reproduction.** A baseline of the form (3.12) – constant in frequency within each zone, times an arbitrary degree- n polynomial in time – must give an identically zero residual at every surviving sample, at any η , under arbitrary masks. Whatever appears is pure accumulated roundoff, and the test asserts against a computed roundoff bound rather than a fixed tolerance (measured, a few percent of it).
- **The Kronecker identity** (3.18). For a window-constant mask, $r_{\min}^{2d} = r_{\min}^{1d}$ to 10^{-10} . This is the test that justifies carrying ϵ over from the pure frequency fit unchanged, and it fails immediately if $\Theta \neq I$ or if the mask family is weakened from window-constant to merely separable.

Beyond those:

- **A dense oracle:** for each (beam, t), build the design matrix Φ of (3.4) explicitly, all $((2W+1)N_{\text{freq}}) \times (N_\phi(n+1))$ of it, row-select it by the mask, and solve with dense linear algebra. It shares no code with the fast path below the basis tables – no window moments, no banded layout, no coefficient-major ordering – so agreement (measured 4×10^{-15}) is a genuine cross-check of the stencil, the assembly and the banded factorization rather than a restatement of them.
- **The $n = 1$ degeneracy, asserted from both sides.** For a window-constant mask the $n = 1$ fit must equal the $n = 0$ fit exactly (measured 2×10^{-16}), since the odd moments vanish by (3.10); and it must *differ* where the mask varies within the window (measured 0.97), which is what keeps the first half from passing vacuously and is the only evidence that $n > 0$ earns its keep.

- **The rank condition** of §3.4: a zone with data at fewer than $n + 1$ distinct offsets must be flagged, with a positive control that exactly $n + 1$ offsets survives.
- **The bandwidth** (3.11), measured from the assembled matrix. This is a memory-safety test rather than an accuracy one: the $\max(n_\phi, 1)$ binds only at $n_\phi = 0$, where getting it wrong writes out of bounds instead of producing a wrong answer.
- **Chunk invariance**, bit-identical across chunkings; and **masked data unread**, by poisoning flagged samples with NaN and requiring bit-identical outputs.
- **Float32 against float64**, each at its own ϵ – deliberately differing by 10^3 , so that the comparison is made on the intersection of two *different* masking decisions rather than on samples the two runs were guaranteed to agree about. The cumulative mask-expansion rate is tracked alongside, since a systematic bias in float32’s $r_{\min}$ would show up there rather than as a residual discrepancy.

Everything is parameterized over n_ϕ and (n, W) rather than run at one configuration. That is not thoroughness for its own sake: $n_\phi = 0$ is where the regulator is wider than the data block, and the bandwidth bug it exposes is invisible at every other degree.

Generating masks, which is most of the work. The central fact is that *random masks do not find the hard cases*. Measured at $n_\phi = 2$, $K = 5$, $h = 3000$, $\eta = 10^{-3}$: five thousand Bernoulli masks at log-uniform density reach a worst $r_{\min}$ of 9.6×10^{-3} , while a constructed adversarial mask reaches 4.1×10^{-4} – a factor of 23. A suite built on Bernoulli masks would report an enormous margin over ϵ no matter how wrong the conditioning handling was.

The worst case can be constructed exactly, because of a structural accident worth recording. Fix a coefficient vector α and let $w_{fs} \in \{0, 1\}$ be the mask. The equilibrated Rayleigh quotient is *linear-fractional* in w :

$$R(w) = \frac{\sum_{f,s} w_{fs} a_{fs} + A_0}{\sum_{f,s} w_{fs} c_{fs} + B_0}, \quad a_{fs} = b(f, s)^2, \quad c_{fs} = \sum_{jq} \alpha_{jq}^2 \phi_j(f)^2 \psi_q(s)^2, \quad (3.22)$$

with $A_0 = \eta \alpha^T \mathcal{R} \alpha$ and $B_0 = \eta \sum_{jq} \alpha_{jq}^2 (D_1)_{jj} \Theta_{qq}$ *mask-independent* – precisely because the regulator never sees the mask (§3.3). So at level τ the minimizing mask is exactly $\{(f, s) : a_{fs} < \tau c_{fs}\}$, and Dinkelbach’s algorithm converges to the exact minimizer of R over all $2^{N_{\text{freq}}(2W+1)}$ masks at fixed α ; alternating that with $\alpha \leftarrow$ (bottom eigenvector) gives the adversarial mask. Verified two ways: the decomposition reproduces the true Rayleigh quotient to 6×10^{-16} , and no single (f, s) flip improves the converged mask in 40 of 40 configurations.

The generator is then a mixture, drawn independently per zone: the adversarial construction; structured families taken from the measured extremal geometry and from the code’s corner cases (dead zones, leading versus trailing dead runs, $M_z = 0, 1, 2$, combs, whole knot blocks); about 10% separable products of two 1-d masks; and plain Bernoulli for breadth. The result is then perturbed by a random set of fully-flagged and fully-valid rectangles whose extents follow a power law, which is what finds problems nobody predicted. The knot vector is randomized too and targets h *directly* rather than hoping a uniform draw produces a wide interval – including the $K = 1$ case where the whole band is a single knot interval, which by (3.19) is the extreme.

A Fourier-space response of the detrenders

This appendix collects, in one place and in closed form, the frequency response of the detrenders described above, together with CHIME’s — and explains why the last of those needs two functions rather than one. Figure 1 plots all five. The script that draws it, `notes/figs/plot_detrender_response.py`, evaluates only the formulas below and imports nothing from `pirate_frb`.

A.1 Definition, for a stationary detrender

Write the detrender’s output as the residual $r = d - \hat{f}$. For a fixed mask the estimator is linear (§1), $\hat{f} = Hd$, so $r = (I - H)d$. On a *fully valid* mask, and away from the ends of the array, both 1-d detrenders in these notes are additionally time-invariant: H is then a convolution with an *equivalent kernel* h , $\hat{f}[t] = \sum_s h[s] d[t - s]$, and everything about the detrender is contained in one function of frequency,

$$\hat{r}(\omega) = F(\omega) \hat{d}(\omega), \quad F(\omega) = 1 - H(\omega), \quad H(\omega) = \sum_s h[s] e^{-i\omega s}. \quad (\text{A.1})$$

The time coordinate is the full integer line, so $-\pi \leq \omega \leq \pi$.

Note the sign convention, which is the opposite of the one usually wanted for a smoother: F is the response of the *detrender*, so it is a high-pass. $F = 1$ means a sinusoid at that frequency passes through untouched; $F = 0$ means it is entirely absorbed into the baseline and does not reach the search. Every detrender here reproduces constants exactly, so $\sum_s h[s] = 1$ and $F(0) = 0$.

Two properties follow immediately from the kernel. $h[0]$ is the leverage (1.2) of the fully-valid window, so the flux a narrow pulse loses to the baseline is $h[0]$ and the variance the detrender removes is $h[0]$. And if h is symmetric then F is real: the two local polynomial fits evaluate at the center of a symmetric window, so they have exactly zero phase. The fixed-lag Kalman filter does not — truncating the future at lag L breaks the symmetry — but the effect is small, under 1.3° wherever $|F| > 0.1$ at $L = 384$.

A.2 Closed forms

Local polynomial subtraction. At the center of a fully valid window of $2W + 1$ samples the fit is $\hat{f}[t] = \sum_s h[s] d[t + s]$ with $h[s] = \sum_j (M^{-1})_{0j} s^j$ on $|s| \leq W$, where $M_{jl} = \sum_{s=-W}^W s^{j+l}$ is the unweighted moment matrix of (1.6). Odd moments vanish, so M decouples into even and odd blocks and the kernel is even. Writing $N = 2W + 1$, $P = W(W + 1)$, and

$$D_W(\omega) = \sum_{s=-W}^W e^{-i\omega s} = \frac{\sin(N\omega/2)}{\sin(\omega/2)} \quad (\text{A.2})$$

for the Dirichlet kernel (with $D_W(0) = N$):

- $n = 1$: the odd basis function contributes nothing at the window center, so the equivalent kernel is *exactly* the boxcar, $h[s] = 1/N$. ($n = 0$ and $n = 1$ therefore have the same kernel on a full mask, and differ only once a mask makes the window asymmetric.) Hence

$$F(\omega) = 1 - \frac{D_W(\omega)}{N} = \frac{W(W + 1)}{6} \omega^2 + O(\omega^4). \quad (\text{A.3})$$

- $n = 2$: the kernel is $h[s] = a + bs^2$ with

$$a = \frac{3(3P - 1)}{N(4P - 3)}, \quad b = \frac{-15}{N(4P - 3)}, \quad \text{so} \quad h[0] = a \rightarrow \frac{9}{8W}, \quad (\text{A.4})$$

which is the leverage (1.12) of a fully valid window, as it must be. With $S_W(\omega) = \sum_s s^2 e^{-i\omega s} = -D_W''(\omega)$, evaluated as

$$S_W(\omega) = \frac{N^2 + 1}{4} D_W(\omega) + \frac{N \cos(N\omega/2) \cos(\omega/2) - D_W(\omega)}{2 \sin^2(\omega/2)}, \quad (\text{A.5})$$

the response is

$$F(\omega) = 1 - a D_W(\omega) - b S_W(\omega) = \frac{(W-1)W(W+1)(W+2)}{280} \omega^4 + O(\omega^6). \quad (\text{A.6})$$

The two low-frequency limits are the useful summary: a degree- n fit reproduces polynomials of degree $\leq n$, and by symmetry also degree $n+1$ when n is even, so F vanishes as ω^2 at $n=1$ and ω^4 at $n=2$. A steeper rolloff attenuates *harder* below the corner frequency: at $\omega = 10^{-3}$ the $n=1$ curve of Figure 1 sits two orders of magnitude above the $n=2$ one, meaning it is the *weaker* high-pass and leaves correspondingly more trend behind.

Kalman filter. Already derived as (2.15): in the two-sided limit the normal equations diagonalize in Fourier space, giving

$$F(\omega) = \frac{\rho (2 \sin(\omega/2))^{2k}}{1 + \rho (2 \sin(\omega/2))^{2k}}, \quad \rho = \tau^{2k}, \quad (\text{A.7})$$

which is $(\tau\omega)^{2k}$ at low frequency and, unlike the local fits, is monotone in $|\omega|$ with no stopband ripple. Figure 1 plots this $L \rightarrow \infty$ form. At the finite lookahead actually used, $L = 384 = 4.2 \ell$, the true response differs from it by at most 6×10^{-3} in absolute value and about 4% in the passband — invisible at the scale of the plot.

A.3 The CHIME detrender, for which F is not one function

CHIME's real-time pipeline detrends in time with the `polynomial_detrender` of `rf_pipelines`, configured in `ch_frb_l1` as `polydeg=4`, `nt_chunk=1024`, `axis=AXIS.TIME`. It fits Legendre polynomials $P_0 \dots P_4$ by weighted least squares over each disjoint $n_t = 1024$ sample block of a *fixed* lattice, and subtracts the fit in place. Take uniform weights, ignore the conditioning cut that masks a badly conditioned block, and apply it once; then

$$R = I - \Pi, \quad \Pi = \bigoplus_{\text{blocks}} Q Q^T, \quad (\text{A.8})$$

with Q an orthonormal basis for degree ≤ 4 on one block. Π and hence R are *orthogonal projections*: $R^2 = R$ and $R^T = R$. Neither detrender of these notes has that property — a sliding window fit is a projection row by row but not globally, and a penalized estimator shrinks rather than projects.

The difficulty is that R commutes with a shift by n_t but not by one sample. It is linear *periodically* time-varying, not time-invariant, so (A.1) does not apply: an operator that is block-diagonal in time is dense in frequency, and a sinusoid comes back as a comb,

$$R e_\omega = \sum_{k=0}^{n_t-1} A_k(\omega) e_{\omega+2\pi k/n_t}, \quad e_\omega[t] \equiv e^{i\omega t}. \quad (\text{A.9})$$

The complete description is the two-index object $A_k(\omega)$, and any scalar $F(\omega)$ is a projection of it onto one index — so choosing a definition means choosing a question.

The block lattice is set by the acquisition clock and has nothing to do with when a burst arrives, which makes it natural to average over its phase. Doing so restores stationarity, and leaves exactly two second-order summaries: $\mathbb{E}[R]$ is time-invariant with symbol $A_0(\omega)$, and $\mathbb{E}[R^T R]$ is time-invariant with symbol

$\sum_k |A_k(\omega)|^2$. Hence

$$F_{\text{coh}}(\omega) = |A_0(\omega)|, \quad F_{\text{pow}}(\omega) = \left(\sum_k |A_k(\omega)|^2 \right)^{1/2}. \quad (\text{A.10})$$

F_{coh} is the amplitude that survives *at the input frequency*, which is what a narrowband filter at ω would see coherently; F_{pow} is the total surviving power, at any frequency. Their difference,

$$F_{\text{scat}} = (F_{\text{pow}}^2 - F_{\text{coh}}^2)^{1/2}, \quad (\text{A.11})$$

is power the detrender moves to a *different* frequency, and is identically zero for any time-invariant detrender: it exists only because of the block seams.

Both reduce to a single block, which is what makes them cheap to evaluate. Writing $u[n] = e^{i\omega n}$ on $n = 0 \dots n_t - 1$ and matching (A.9) block by block gives $A_k(\omega) = \text{FFT}_k(((I - \Pi)u)[n]e^{-i\omega n})/n_t$, hence

$$F_{\text{coh}}(\omega) = 1 - \frac{\|Q^T u\|^2}{n_t}, \quad F_{\text{pow}}(\omega)^2 = \frac{\|(I - \Pi)u\|^2}{n_t}. \quad (\text{A.12})$$

These are what the script evaluates: one $n_t \times 5$ product per frequency, exact at any ω .

The two are not independent. Because R is an orthogonal projection, $R^T R = R^2 = R$, so

$$F_{\text{pow}}^2 = \frac{u^\dagger R^\dagger R u}{n_t} = \frac{u^\dagger R u}{n_t} = F_{\text{coh}}, \quad \text{and therefore} \quad F_{\text{scat}} = \sqrt{F_{\text{coh}}(1 - F_{\text{coh}})} \leq \frac{1}{2}. \quad (\text{A.13})$$

So there is really one function and a square, which is a happier situation than it first appears: one does not have to adjudicate between the two definitions, only to say which is being quoted. Three consequences are worth recording. The worst-case scattering is exactly 1/2 in amplitude and occurs exactly at the coherent half-power point, which for this configuration is $\omega = 0.0122$ (a period of 516 samples); there, half the input amplitude emerges somewhere else. That scattered power is not a smooth smear but a comb of spacing $2\pi/n_t = 0.0061$, comparable to the corner frequency itself. And because squaring doubles a slope on a log-log plot, the two curves diverge badly in the passband: a degree-4 fit reproduces $t^0 \dots t^4$, so $F_{\text{pow}} \propto \omega^5$ while $F_{\text{coh}} \propto \omega^{10}$.

A.4 What the figure shows

The three configurations plotted are matched at their half-power points, $\omega_* = 0.0148, 0.0156$ and 0.0156 respectively, which is the only way a comparison of this kind means anything. Two features are worth reading off.

The $n = 2$ fit and the Kalman filter are nearly the *same filter* over the whole band, despite arriving there by quite different routes; both roll off as ω^4 , so matching them anywhere in the passband nearly matches them everywhere in it. Where they differ is the stopband: the local fit is a truncated polynomial kernel and rings, reaching $|F| = 1.24$ near $\omega \approx 2.2\pi/W$ — a band in which it *amplifies* — while the Kalman response is monotone by inspection of (A.7). The $n = 1$ fit is a different animal: matched at the same half-power point, its shallower ω^2 rolloff leaves it an order of magnitude higher through the passband.

For CHIME, no single number can be quoted as “the response” in the passband. At a period of 1500 samples the two definitions differ by a factor of 81, and they fall on *opposite sides* of the $n = 2$ curve: $F_{\text{coh}} = 1.5 \times 10^{-4}$, the local fit 4.6×10^{-3} , $F_{\text{pow}} = 1.2 \times 10^{-2}$. The half-power point moves too, from $\omega = 0.0122$ under F_{coh} to 0.0105 under F_{pow} , a ratio of 1.16. A block detrender is simply not summarized by a transfer function the way a sliding one is, and the gap between these two curves is a fair measure of how much is being left out.

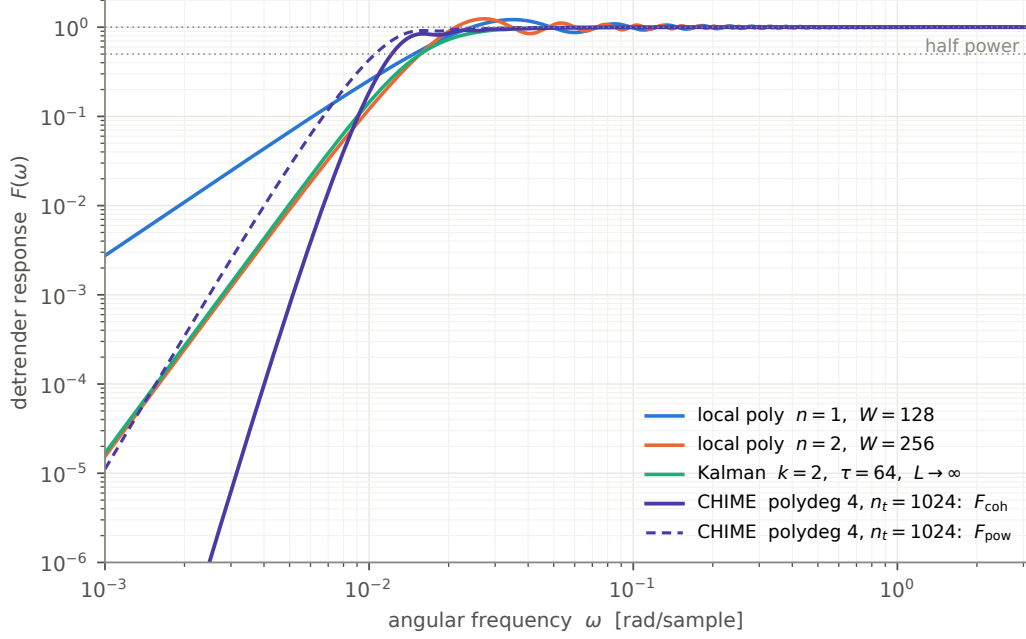


Figure 1: Fourier-space response $F(\omega)$ of five detrender configurations, all on a fully valid mask. F is the response of the *residual*: $F = 1$ means a sinusoid survives untouched, $F = 0$ means it is absorbed into the baseline. The three stationary curves are (A.3), (A.6) and (A.7); the Kalman curve is the $L \rightarrow \infty$ limit, which at the $L = 384$ actually used is accurate to 6×10^{-3} . The CHIME detrender is not time-invariant, so it contributes two curves, (A.10): F_{coh} , the amplitude surviving at the input frequency, and F_{pow} , the total surviving power at any frequency. They are related by $F_{\text{coh}} = F_{\text{pow}}^2$, (A.13). Generated by `notes/figs/plot_detrender_response.py`.